



On the incidence of bank levies: theory and evidence

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Abstract

In the aftermath of the financial crisis, several European countries have introduced levies on bank liabilities. The aim is to compensate taxpayers for the provision of bailouts and guarantees and to internalize the fiscal costs of future banking crises. This paper studies the tax incidence: Building on the Monti–Klein model, we predict that banks shift the tax mainly to borrowers by raising lending rates and that deposit rates may increase because deposits are partly exempt. Bank-level evidence from 23 EU countries (2007–2013) shows that the levy indeed increases the lending and the deposit rate as well as the net interest margin. Banks adjust differently to this tax depending on the composition of their balance sheets: In line with theory, especially those banks with a high loan-to-deposit ratio raise the interest rates. Market concentration and the capital structure influence the magnitude of the pass-through, which is stronger in concentrated markets and weaker in case of banks with a high regulatory capital ratio.

Keywords Bank levy · Taxation of banks · Tax incidence · Pigovian taxes

JEL Classification G21 · G28 · H22

1 Introduction

Banking crises are expensive: Beyond their impact on the real economy, they often involve large fiscal costs as a substantial amount of public funds is spent on the stabilization of the banking sector. During the financial crisis, EU member states, for example, incurred a fiscal cost for bank recapitalization and asset relief (2008–2013) of 4.9% of GDP (European Commission 2013). In addition, government guarantees

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and liquidity assistance were provided that in 2009 reached a peak value of 6.9% of GDP. There are, however, large differences in those magnitudes across countries: While they were rather moderate in Germany or Austria, the UK incurred costs of recapitalization and guarantees worth 7.4 and 10.1% of GDP, respectively. In Ireland, they even reached 39.9 and 173.8% of GDP.

Following a request of the G-20, the IMF issued a report about special taxes on banks with the aim of ‘fair and substantial contribution by the financial sector.’ The objective is to (i) raise revenue to cover the fiscal costs of (future) banking crises, thereby compensating taxpayers for guarantees and bailouts and (ii), as a Pigovian tax, to internalize externalities associated with such guarantees, thereby enhancing financial stability and complementing regulation. Since 2009, 15 European countries including Germany and the UK have introduced such levies typically as a tax on bank liabilities.

A central aspect of every tax is the incidence, which is informative about the distributional consequences. In this case, knowing whether banks and their owners bear the tax themselves or shift it to others is of particular interest. After all, raising a contribution from those who benefited the most from guarantees and bailouts was one of the reasons why bank levies were introduced in the first place.

This paper both theoretically and empirically analyzes the incidence of bank levies in Europe. The main question is whether banks can shift the tax to their customers by increasing lending or lowering deposit rates. The focus on a pass-through to borrowers and depositors is motivated by its economic relevance as well as by the empirical finding of Goodspeed (2014) that the incidence on wages is generally of minor importance in banking as opposed to manufacturing. Building on the Monti–Klein framework, we develop several scenarios for the tax incidence, thereby carving out potential determinants like balance sheet composition, market concentration, and capital structure. Subsequently, the predictions are taken to the data: Using a cross-country panel dataset with financial information of 2776 EU banks (2007–2013), the effect of the levy on lending and deposit rates as well as on the net interest margin is estimated. All results are derived from a comparison of banks with a similar approximate capital structure in countries that were hit by the financial crisis in a similar way, which accounts for specific shocks due to the crisis and subsequent regulatory reforms.

We find that banks shift part of the tax to borrowers, while depositors even tend to benefit from a higher interest rate because deposits are partly exempt. The average effects are moderate: For example, the lending rate and the net interest margin with average values of 5.86 and 2.49% only increase by 0.24 and 0.05 percentage points, respectively, whenever a bank is taxed. In line with theory, we find empirical evidence for heterogeneous responses: The initial composition of balance sheets is crucial as banks adjust differently depending on the ratio between loans and customer deposits. Especially, ‘loan-rich, deposit-poor’ banks mainly adjust at the loan margin and pass the tax onto borrowers. In addition, the incidence on borrowers is strong and economically significant in highly concentrated banking markets. To a lesser extent, it also depends on the regulatory capital ratio as well-capitalized banks are less exposed to a tax on liabilities.

This work builds on the tax incidence literature in banking. On the theoretical side, Caminal (2003) developed a taxation model with banks and analyzed the effects of a

variety of taxes. He stresses the importance of the separability of loans and deposits and the role of market power. Bierbrauer (2014) studied the incidence of the proposed financial transactions tax on financial markets in a model with fire sales. However, none of these contributions specifically addresses bank levies. On the empirical side, two papers analyze the incidence of bank levies at the country level: Buch et al. (2016) studied the levy in Germany using a difference-in-difference approach that exploits the exemption of small banks. They find a decrease in the loan volume but no effect on the lending rate. Instead, the levy increases the deposit rate. The latter can be explained by the exemption of customer deposits, which induces banks to shift the funding sources toward deposits. For Hungary that introduced a tax on loans, Capelle-Blancard and Havrylchyk (2017) estimated an increase in the lending rate. Motivated by the Monti–Klein model, they argue that the magnitude of the pass-through depends on market power and find evidence for this prediction: The incidence is especially on those customers with an existing borrowing relationship, who thus face a high cost of switching to another bank.

A related strand analyzes the incidence of the corporate income tax on banks: Demirciğ-Kunt and Huizinga (1999) found evidence that it is fully passed onto customers as the net interest margin increases one by one with the tax rate. In the same spirit, Demirciğ-Kunt and Huizinga (2001) argued that the pass-through is weaker in case of international banks that can exploit profit shifting opportunities and are less sensitive to domestic taxes. Albertazzi and Gambacorta (2010) showed that the corporate income tax raises the lending rate such that up to 90% of the tax is passed onto borrowers, which is consistent with Chiorazzo and Milani (2011) who estimated a pass-through of 45% in the short and 80% in the long run. Using a different measure of the bank's tax liability, Capelle-Blancard and Havrylchyk (2014), however, found no evidence for a pass-through.

We add to this literature a study of the incidence of bank levies that combines theory and empirical evidence. This approach reveals the main adjustment margins of banks, rationalizes heterogeneous and potentially countervailing effects, and quantifies the pass-through. Compared to two prior studies on bank levies, we extend the analysis and present cross-country evidence. This approach commonly used in the literature on the incidence of the corporate income tax yields more general insights. Importantly, with the recent establishment of the Banking Union, Euro area banks have to contribute to a single resolution fund. These contributions share some similarities with the bank levy; for example, they are charged on bank liabilities. Cross-country evidence on the previously introduced national levies can thus be informative about the incidence of these new contributions. The present paper also highlights two novel heterogeneous effects of the bank levy, which can be rationalized by the model: First, the magnitude of the pass-through to borrowers does not only depend on market concentration but also on the capital structure of banks. Second, the composition of balance sheets determines whether banks mainly adjust at the loan or deposit margin. This importantly influences the tax incidence.

Eventually, this paper connects to the literature on Pigovian taxes in banking, which offers an explicit motivation of bank levies. In the aftermath of the financial crisis, such taxes have been proposed to internalize externalities associated with the collapse and bailout of banks (Keen 2011), short-term debt financing (Perotti and Suarez 2011),

and systemic risk and undercapitalization (Acharya et al. 2017). Devereux et al. (2015) evaluated bank levies in Europe and their effects on banks' capital structure and risk taking: They find evidence that the levy has the intended effect on the capital structure because banks increase equity and reduce leverage. However, they also take more risks measured by the average risk weight. The latter reflects a mechanical effect as additional equity allows for higher risk-weighted assets, and banks typically adjust at the risk margin.

The remainder of this paper is organized as follows: Sect. 2 provides an overview about bank levies in Europe. Section 3 sets out the model and derives the main predictions about the incidence, which are taken to the data in Sect. 4. Finally, Sect. 5 concludes.

2 Bank levies

In 2010, the IMF published a report on the taxation of the financial sector¹ and proposed a bank levy, the so-called financial stability contribution. It is a tax on the total liabilities of financial institutions excluding equity and deposits covered by a guarantee scheme and targets the risky part of funds like uninsured deposits and wholesale funding. The exemption of insured deposits helps avoid double taxation due to insurance premia. Ideally, the tax rate reflects the funding benefit for large, systemically important banks, which emerges because of implicit government guarantees. According to the IMF's estimates, it averages around 20 basis points.

Since 2009, 11 EU member states have introduced a tax on bank liabilities. Table 1 summarizes the main characteristics in 2013, which marks the end of our observation period. In general, the levies share a common basic design but differ in five aspects: First, the tax rates reach from values below the 20 basis points suggested by the IMF up to 40 basis points. They are either flat (e.g., Belgium, Sweden), progressive (Austria, Germany) or differ between short- and long-term liabilities (Netherlands, UK). Second, four countries exempt small banks by an allowance because the latter are unlikely to benefit from guarantees. Third, the tax base usually consists of total liabilities without equity and insured deposits; Germany even exempts total customer deposits. In general, the tax base is computed using banks' unconsolidated (stand-alone) accounts. The UK and the Netherlands rely on consolidated balance sheets. These two countries also allow for the deduction of selected equity categories (Tier 1 and regulatory capital) only. In addition to liabilities, Austria (until 2014), Germany, and Portugal also charge a levy on derivatives but with a different tax rate. Fourth, the levy is typically imposed on current- or previous-year liabilities except for Austria, which used the 2010 balance sheet as tax base for the years 2011–2013 and switched to the previous-year balance sheet in 2014. Nevertheless, this feature was known to be temporary from the beginning and is unlikely to prevent behavioral responses. Fifth, the revenue either accumulates in a resolution fund (e.g., Belgium, Germany, Sweden), which should help finance the resolution of insolvent banks, or is used for the general budget (e.g., Austria, Netherlands, UK).

¹ 'Financial Sector Taxation. The IMF's Report to the G-20 and Background Material,' 2010.

Table 1 Bank levies in Europe: overview. *Sources:* OECD (2013), Devereux et al. (2015, Appendix)

	Tax base	Tax rates	Exemptions
<i>Austria</i> 1.1.2011	Total liabilities	< EUR 20bn: 0.055% > EUR 20bn: 0.085%	Insured deposits Allowance: EUR 1bn
<i>Belgium</i> 1.1.2012	Total liabilities	0.035%	Insured deposits
<i>Cyprus</i> 1.10.2011	Total deposits	0.011%	Interbank deposits
<i>Germany</i> 1.1.2011	Total liabilities	< EUR 10bn: 0.02% EUR 10bn–100bn: 0.03% EUR 100bn–200bn: 0.04% EUR 200bn–300bn: 0.05% > EUR 300bn: 0.06%	Customer Deposits Allowance: EUR 300m Cap: 20% of net income Minimum charge*: 5%
<i>Latvia</i> 1.10.2011	Total liabilities	0.072%	Insured deposits
<i>Netherlands</i> 1.10.2012	Total balance sheet	0.044% (short-term) 0.022% (long-term)	Insured deposits Regulatory capital Allowance: EUR 20bn
<i>Portugal</i> 1.10.2011	Total liabilities	0.05%	Insured deposits
<i>Romania</i> 2.6.2011	Total liabilities	0.1%	Insured deposits
<i>Slovakia</i> 1.1.2012	Total liabilities	0.4%	Insured deposits, Subordinated Debt
<i>Sweden</i> 30.12.2009	Liabilities and Provisions	0.044%	Subordinated Debt, Selected Securities
<i>UK</i> 1.1.2011	Total balance sheet	0.036% (short-term) 0.071% (long-term)	Insured deposits, Tier 1 Capital Allowance: GBP 20bn

Information in 2013 (end of observation period). *Only 5% of the tax liability is payable if a bank incurs a loss

In addition to these 11 countries, the empirical analysis also includes 12 EU member states that do not introduce bank levies as a control group (Bulgaria, Czech Republic, Denmark, Estonia, Greece, Ireland, Italy, Lithuania, Luxembourg, Malta, Poland, Spain)

Moreover, four countries adopted conceptually different bank levies. In France, the tax base is minimum regulatory capital, and Hungary and Slovenia impose a levy on bank loans and assets, respectively. Finland adopted a temporary tax on risk-adjusted assets. These levies substantially differ from a tax on liabilities, and we do not include the four countries in the empirical analysis.

As of 2016, financial institutions in all Euro area countries must contribute to the Single Resolution Fund (SRF), which is a key pillar of the newly established Banking Union. The SRF is built up over 8 years and is expected to reach a target volume equal to 1% of insured deposits. Banks' contributions share some similarities with the bank levies; most importantly, they are proportional to total liabilities net of insured deposits. In addition, they are sensitive to a bank's risk profile. These contributions do not automatically replace national bank levies, but some countries modified the tax: Germany completely replaced the levy in 2016. Austria preserved the levy but significantly lowered the tax rates.

3 Theoretical analysis

We apply a variant of the Monti–Klein model with taxes and regulation. This industrial organization model, which goes back to Klein (1971) and Monti (1972), characterizes the lending and borrowing decisions of oligopolistic banks. Despite some controversy—in the neoclassical tradition, the bank is modeled as a banking firm and the only friction is imperfect competition—it features prominently in the tax incidence literature (e.g., Albertazzi and Gambacorta 2010; Capelle-Blancard and Havrylychuk 2017).

This approach is informative about how banks adjust the balance sheets and, most importantly, the interest rates in response to a levy and yields testable predictions about the tax incidence. However, our model is partial equilibrium, and it would not be well suited for exploring potential gains of the levy, for example, in terms of smaller bank leverage and improved financial stability.

3.1 Monti–Klein model with taxation and regulation

Suppose a number of identical banks indexed by $i = 1, \dots, N$ compete for loans and deposits in a Cournot fashion. Each bank provides commercial loans l_i and faces the downward-sloping inverse demand function $r_L = r_L(L)$; $L = \sum_{i=1}^N l_i$ denotes aggregate loans.

A bank can borrow from three different sources: First, it attracts deposits that are protected by a guarantee scheme (insured deposits), s_i , or not (uninsured deposits), d_i . They are characterized by the upward-sloping inverse supply $r_S = r_S(S)$ and $r_D = r_D(D)$ with $S = \sum_{i=1}^N s_i$ and $D = \sum_{i=1}^N d_i$, respectively. Without loss of generality, the insurance premium is normalized to zero. We explicitly picture two distinct markets to study a typical feature of bank levies, namely, the tax exemption of insured deposits. Second, a bank raises an amount m_i from interbank or money markets at a fixed interest rate r determined by central banks or on the international capital market. Whenever m_i is negative, the bank is a net lender with additional assets.² The option of borrowing or lending at a fixed interest rate makes loans and deposits separable. It is a typical property of the Monti–Klein model and influences the tax incidence as pointed out by Caminal (2003). It persists if the bank incurs additively separable administrative costs but is not robust to bankruptcy risk as emphasized by Dermine (1986). Third, a bank can raise equity e_i at a fixed return $\rho > r$. Capital standards require equity to be at least a fraction k of total assets, which consist of commercial loans l_i and, if $m_i < 0$, money market loans.

Each bank chooses loans l_i , insured and uninsured deposits s_i and d_i , the interbank and money market position m_i , and equity e_i to maximize the after-tax surplus of its shareholders

$$\pi_i = (1 + r_L)l_i - (1 + r_S)s_i - (1 + r_D)d_i - (1 + r)m_i - T_i - (1 + \rho)e_i. \quad (1)$$

² Gross and net positions coincide either if loans are subject to capital requirements or liabilities are taxed; see Online Appendix A.3.

T_i denotes the tax liability. The choices must satisfy the balance sheet identity, $l_i = s_i + d_i + m_i + e_i$, and capital requirements, $e_i \geq k[l_i + \min\{m_i, 0\}]$.

The bank levy is a tax on liabilities consisting of customer deposits and—if positive—money market borrowing. Insured deposits s_i are exempt.³ With a uniform tax rate τ , the tax liability of bank i is:

$$T_i = \tau[d_i + \max\{m_i, 0\}]. \quad (2)$$

If banks borrow from money markets, $m_i \geq 0$, balance sheet identity and capital requirements imply $T_i = \tau(m_i + d_i) = \tau[(1 - k)l_i - s_i]$. In this case, the tax increases in loans, is independent of uninsured deposits, and decreases in insured deposits. Hence, substituting the latter for other liabilities allows a bank to lower the tax payment. If banks are money market lenders, $m_i < 0$, in contrast, the tax liability is proportional to uninsured deposits: $T_i = \tau d_i$.

3.2 Equilibrium analysis

We first study the adjustment of ‘loan-rich, deposit-poor’ banks that borrow from money and interbank markets, $m_i \geq 0$. Substituting the tax liability, the balance sheet identity, and the regulatory constraint into (1), yields the consolidated maximization problem:

$$\pi_i = \max_{l_i, d_i, s_i} [r_L - \hat{r} - \tau(1 - k)]l_i + (r + \tau - r_S)s_i + (r - r_D)d_i. \quad (3)$$

$\hat{r} \equiv r(1 - k) + \rho k$ denotes the marginal capital cost of loans. The first-order conditions are $r_L(L) + r'_L(L)l_i - [\hat{r} + \tau(1 - k)] = 0$, $r + \tau - [r_S(S) + r'_S(S)s_i] = 0$, and $r - [r_D(D) + r'_D(D)d_i] = 0$. Optimal loans balance the marginal return and cost; optimal deposits equalize the marginal costs of deposits and the alternative funding source, namely money market borrowing.

In the symmetric equilibrium with $l_i = l = L/N$, $s_i = s = S/N$, and $d_i = d = D/N$, the first-order conditions state that the Lerner index equals the inverse interest rate elasticity of loan demand and deposit supply

$$\frac{r_L - [\hat{r} + \tau(1 - k)]}{r_L} = \frac{1}{N\varepsilon_L}, \quad \frac{r - r_D}{r_D} = \frac{1}{N\varepsilon_D}, \quad \frac{r + \tau - r_S}{r_S} = \frac{1}{N\varepsilon_S} \quad (4)$$

with $\varepsilon_L = -\frac{1}{r'_L(L)} \frac{r_L}{L} > 0$, $\varepsilon_S = \frac{1}{r'_S(S)} \frac{r_S}{S} > 0$ and $\varepsilon_D = \frac{1}{r'_D(D)} \frac{r_D}{D} > 0$. Banks charge a markup on loans and a markdown on uninsured deposits that is inversely related to the elasticity and the number of competitors. As a result, the lending rate exceeds the cost of capital, and the deposit rate falls short of the interbank rate (i.e., $r_L > \hat{r} > r > r_D$).

We differentiate⁴ the equilibrium conditions assuming constant elasticities:

³ If they are not exempt, the levy will affect the respective interest rate r_S in parallel to the interest rate on uninsured deposits r_D .

⁴ We focus on a small tax change such that the regulatory constraint still binds (i.e., $\rho > r + \tau + d\tau$). Otherwise, banks would substitute equity for liabilities.

Proposition 1 *If banks are loan-rich and deposit-poor ($m_i > 0$), the bank levy is passed onto borrowers. It lowers loans L and raises the lending rate:*

$$\frac{\partial r_L}{\partial \tau} = \frac{1 - k}{1 - \frac{1}{N\varepsilon_L}} > 0. \quad (5)$$

The pass-through is stronger if the number of competitors is small and the loan demand inelastic and weaker if banks are subject to tight capital standards.

The bank levy increases insured deposits S and the corresponding interest rate:

$$\frac{\partial r_S}{\partial \tau} = \frac{1}{1 + \frac{1}{N\varepsilon_S}} > 0. \quad (6)$$

The effect is stronger if there are many competitors and the deposit supply is elastic. There is no effect on uninsured deposits D and the respective interest rate r_D .

Proof See Appendix A.1. □

Recall that optimal loans equalize marginal return and cost. When facing a higher borrowing cost due to the levy, banks cut back loans which eventually raises the lending rate. Borrowers thus bear part of the tax. The extent of the pass-through depends on bank competition and capitalization: First, few competitors and an inelastic loan demand *ceteris paribus* reinforce the increase in the lending rate. This is due to market power in the sense that the balance sheet adjustment of one bank has a more pronounced impact on the equilibrium interest rate. Second, the capital structure, dictated by regulation in this model, determines a bank's exposure to a tax on liabilities. If capital requirements are high, banks are less affected by the levy such that the adjustment and the increase in the lending rate are smaller. Proposition 1 also implies that the lending rate usually increases more than proportionately, $dr_L/d\tau > 1$.

In contrast, deposits are unaffected as long as the levy does not change their marginal costs *relative* to other liabilities especially money market borrowing. Put differently, a levy uniformly imposed on total liabilities would not influence the fixed relation between deposit and money market rates. Since insured deposits are exempt, however, they become a relatively cheaper source of funding. Banks thus substitute them for other liabilities, which are fully taxed. The higher demand raises the interest rate on uninsured deposits, thereby even benefiting depositors. It tends to rise by less if the deposit market is concentrated and the supply inelastic. The latter can be attributed to the relative stickiness of deposits, a phenomenon well documented in the empirical literature, for example, by Hannan and Berger (1991).

As a result, the tax is borne by borrowers and shareholders, whose surplus in excess of the competitive return on equity declines.⁵ Money market lenders do not bear the tax because the return is fixed, and depositors benefit from a higher return on the insured part of their deposits.

These findings describe the behavior of banks that are money market borrowers and need additional funds to finance loans. In some markets, however, banks tend to

⁵ The latter follows from an Envelope argument: $\frac{\partial \pi_i}{\partial \tau} = -[(1 - k)l_i - s_i] < 0$.

be net lenders (i.e., $m_i < 0$). Noting (2), the tax liability equals $T_i = \tau d_i$, and capital requirements are based on both commercial and money market loans $e_i \geq k(l_i - m_i)$. Substituting for m_i , e_i , and T_i in (1) gives:

$$\pi_i = \max_{l_i, d_i, s_i} (r_L - r)l_i + (\tilde{r} - r_D - \tau)d_i + (\tilde{r} - r_S)s_i \quad (7)$$

with $\tilde{r} \equiv (r - \rho k)/(1 - k)$ denoting the net return on one unit of deposits invested on the money market.⁶ The conditions for the symmetric equilibrium are:

$$\frac{r_L - r}{r_L} = \frac{1}{N\varepsilon_L}, \quad \frac{\tilde{r} - (r_D + \tau)}{r_D} = \frac{1}{N\varepsilon_D}, \quad \frac{\tilde{r} - r_S}{r_S} = \frac{1}{N\varepsilon_S}. \quad (8)$$

Differentiating these conditions establishes:

Proposition 2 *If banks are loan-poor and deposit-rich ($m_i < 0$), the levy is passed onto depositors. It lowers uninsured deposits D and the respective deposit rate:*

$$\frac{\partial r_D}{\partial \tau} = -\frac{1}{1 + \frac{1}{N\varepsilon_D}} < 0. \quad (9)$$

The pass-through is weaker if there are few competitors and the deposit supply is inelastic.

The bank levy neither affects loans L and the lending rate r_L nor insured deposits S and the respective interest rate r_S .

Proof See Appendix A.1. □

Compared to the first case summarized in Proposition 1, the results are reversed. The levy does not affect the optimal asset allocation between commercial and money market loans. Therefore, banks do not reduce loans and the tax is not shifted to borrowers. On the borrowing side, the marginal costs of deposits must match the fixed return on assets $\tilde{r}$. The levy mechanically raises the cost of uninsured deposits, banks reduce their demand, and the respective interest rate r_D falls. If banks are money market lenders, the tax incidence is on depositors and shareholders, whereas borrowers are unaffected.

The analysis reveals two scenarios—essentially a pass-through to borrowers or depositors—depending on banks' balance sheet composition. This distinction is a result of the separability of loans and deposits: If banks did not participate in the money market, loans and deposits would be connected by a funding constraint and the levy would eventually increase the lending rate and decrease the rate on uninsured deposits (see Online Appendix A.2). As soon as banks can also borrow and lend at a constant rate, either the deposit or the lending rate is fixed.

Interbank and money markets clear on a worldwide scale. Given that banks in one market are similar in a symmetric equilibrium, they are mainly interbank borrowers

⁶ One unit of deposits creates $1/(1 - k)$ units of assets, which earn $r/(1 - k)$ on the money market. To satisfy capital standards, the bank must raise $k/(1 - k)$ units of equity with return ρ .

in some markets and lenders in others. The incidence thus differs depending on where the bank levy is introduced. Our first scenario with a rising lending and deposit rate applies in markets with a strong and elastic loan demand, a fragmented loan market, or a low money market rate. These factors lead to banks with large loans relative to deposits (see (4)). We consider this case the main scenario because the richer liability structure is more suitable for analyzing the levy, which specifically targets the risky part of bank funding. Most importantly, the main empirical findings are consistent with this scenario. Decreasing deposit rates is also unlikely in times when interest rates are close to the zero lower bound.

In general, the model suggests a linear effect of the levy on interest rates such that an increase in the tax rate leads to the same response irrespective of the initial level (see (5), (6), and (9)). In part, this property can be attributed to the assumption of constant demand and supply elasticities. Online Appendix A.1 relaxes this assumption and allows for a more general loan demand with a variable elasticity. In this case, the relation between tax and lending rates can become concave for quite typical demand functions.

Eventually, we analyze how the bank levy affects the net interest margin (NIM) that represents a main outcome variable in the empirical analysis. It measures the interest rate spread defined as net interest income divided by interest-bearing assets:

$$\text{NIM} = \frac{r_L l_i - r m_i - r_S s_i - r_D d_i}{l_i - \min\{m_i, 0\}}. \quad (10)$$

Substituting for money market borrowing m_i yields

$$\text{NIM} = \begin{cases} r_L - r(1-k) + \frac{(r-r_S)S + (r-r_D)D}{L}, & \text{if } m_i \geq 0 \\ \frac{(1-k)[(r_L-r)L + (\tilde{r}-r_S)S + (\tilde{r}-r_D)D]}{D+S}, & \text{if } m_i < 0 \end{cases} \quad (11)$$

which uses the funding constraint as well as $d_i/l_i = D/L$, $s_i/l_i = S/L$, and $l_i/(d_i + s_i) = L/(D + S)$ in the symmetric equilibrium. If banks are money market lenders, interest-bearing assets consist of commercial and money market loans $l_i - m_i$ and are equal to total funds $s_i + d_i + e_i = (s_i + d_i)/(1 - k)$. In both scenarios, we find:

Proposition 3 *Introducing a bank levy increases the net interest margin.*

Proof See Appendix A.1. □

The increase is driven by a higher lending rate r_L and smaller loans L in the main scenario and by a higher interest rate spread on uninsured deposits $\tilde{r} - r_D$ and a smaller deposit volume, otherwise. Although the responses are both positive, their magnitude may differ.

3.3 Predictions

The theoretical analysis reveals two scenarios, which are summarized in Table 2.

In the *main scenario* (upper panel) describing markets with relatively loan-rich, deposit-poor banks (Proposition 1), the incidence is on borrowers. Banks also exploit

Table 2 Theoretical predictions

		Marginal effect of levy	Influence of	
			Concentration	Capital ratio
<i>Main scenario</i>				
Lending rate	+	↑	↓	
Deposit rate	+	↓	None	
NIM	+	Ambiguous	Ambiguous	
<i>Alternative scenario</i>				
Lending rate	None			
Deposit rate	—	↓	None	
NIM	+	Ambiguous	Ambiguous	

This table summarizes the predictions about the levy's effect on interest rates and NIM and indicates and whether it is reinforced (↑) or weakened (↓) by high market concentration and a high capital ratio in case of loan-rich, deposit-poor (upper panel) and loan-poor, deposit-rich banks (lower panel)

the tax exemption and attract additional deposits, which leads to a higher interest rate at least on the uninsured part of deposits. We cannot distinguish between interest paid on the insured and uninsured part in the data and thus expect an increase in the average deposit rate. Market concentration and bank capitalization influence the magnitude of the pass-through.

The *alternative scenario* (lower panel) summarizes the adjustment to the bank levy in markets with mainly loan-poor, deposit-rich banks (Proposition 2). Banks shift the tax to depositors by offering a lower interest rate on uninsured deposits, while the rate on insured deposit remains constant. The average deposit rate thus tends to fall. The lending rate is unchanged and borrowers are not affected.

4 Empirical evidence

This section provides empirical evidence about the incidence of bank levies. We conduct reduced-form tests based on the theoretical predictions and employ a panel dataset from Bankscope with balance sheet data of 2776 banks in 23 EU member states between 2007 and 2013.⁷

4.1 Estimation strategy

4.1.1 Baseline model

The main econometric specification is

$$y_{ijt} = \alpha_i + \gamma_t + \beta_1 \text{Levy}_{ijt} + \varphi X_{ijt} + \varepsilon_{ijt} \quad (12)$$

⁷ This *de facto* captures the (post-)crisis period 2008–2013 as bank-level stock variables are lagged.

with indices i , j , and t for bank, country, and year, respectively. y_{ijt} and Levy_{ijt} are outcome and main explanatory variable, α_i and γ_t represent bank and year fixed effects and $\mathbf{X}_{ijt}$ the vector of controls.

The analysis focuses on three outcome variables: the interest income on loans as a share of average loans, IIL_{ijt} , the interest expense on customer deposits as a share of average customer deposits, IED_{ijt} , and the net interest margin, NIM_{ijt} . The first two variables equal the average interest rates paid by borrowers and to depositors and approximate lending and deposit rates that are not available in the data. They are conservative measures and may underestimate the magnitude of the pass-through because one cannot distinguish between interest rates for old and new contracts, for example. Moreover, defaults influence the interest income on loans, and the pass-through to good borrowers is likely more pronounced.⁸ The net interest margin—the ratio of net interest income to average interest-bearing assets—is included as a measure of the interest rate spread.

We represent a bank's tax exposure, Levy_{ijt} , by three proxies that exploit different sources of variation, namely (i) only some countries adopt a levy, (ii) some countries exempt small banks, and (iii) marginal tax rates differ across and in case of a progressive tax also within countries. The country-year dummy variable Levy1_{jt} indicates whether a bank is located in a country that charges a levy in a certain year. It measures the effect on average banks but may be affected by other country-year level variations such as changes in corporate taxes. The bank-year dummy variable Levy2_{ijt} indicates whether a bank is effectively taxed to account for the fact that Austria, Germany, the Netherlands, and the UK exempt small banks. This measure equals one if both a levy is charged in a year and the taxable liabilities of a bank (see Sect. 4.2) exceed the allowance. In all other countries, Levy2 is the same for all banks. We also include the statutory marginal tax rate Levy3_{ijt} . If the bank levy is progressive, we choose the marginal tax rate that applies for a bank according to its taxable liabilities; otherwise, Levy3 is the same for all banks that are taxed.⁹ As suggested by Albertazzi and Gambacorta (2010), the regression also includes the quadratic term Levy3^2 to account for potential nonlinearities. Interest rates can be nonlinear functions of the tax once we rely on more general specifications of loan demand and deposit supply with variable instead of constant elasticities. More specifically, Online Appendix A.1 sets out such an example and demonstrates that the lending rate is concave in the tax rate if firms with a standard production function finance only part of their investment with bank loans.

The bank-level tax variables Levy2 and Levy3 depend on balance sheet characteristics and are considered endogenous. For instance, a withdrawal of funds, which is part of the error term, affects both the interest expense and the tax exposure. Or, banks around the allowance threshold might strategically lower their taxable liabilities to avoid the levy. This problem is addressed by instrumenting those variables: We apply a methodology developed by Gruber and Saez (2002) to estimate the elasticity of taxable income, which is well established in the public economics literature and also

⁸ This holds a fortiori if a higher lending rate as a result of the levy increases risk taking and defaults in the sense of Stiglitz and Weiss (1981).

⁹ Table 15 in Appendix A.2 provides an overview about the marginal tax rates.

applied in a related paper of Devereux et al. (2015). The instrument is the bank's predicted exposure to the levy, which is calculated using its balance sheet prior to the levy's introduction. To avoid any confounding effects of the financial crisis, which influenced balance sheets and motivated the bank levy, we rely on the pre-introduction and pre-crisis balance sheet of 2007 to compute the predicted exposure.¹⁰ Predicted and actual exposures are highly correlated with correlation coefficients of 0.89 (Levy2) and 0.98 (Levy3), and the former is a valid potential instrument. The first stage is very strong and the usual F-statistic always clearly exceeds ten. The instrument must also satisfy the exclusion restriction: Although past balance sheets possibly affect interest rates especially if the latter are averages, the instrument is a predicted, hypothetical exposure which—given that we control for bank size and (indirectly) capital structure—would have no own effect if the levy was not actually adopted. Thus, it is plausible that the predicted exposure only influences interest rates through the actual exposure.

A major challenge for identification is the financial and the Eurozone crisis and the government and central bank interventions during the sample period. In addition, regulatory reforms were put in force like the capital requirements directives CRD II and III (effective 2010/11) or proposed like CRR/CRD IV ('Basel III') and the Banking Union (effective 2014/16). These events coincide with the adoption of bank levies and likely affect interest rates. In particular, losses during the crisis and regulatory reforms may cause deleveraging, which eventually raises the lending rate but hardly accounts for a higher deposit rate. Year fixed effects may not adequately capture these developments because new regulations affect banks differently.

This concern is addressed by including separate year fixed effects γ_t for different groups of banks, which absorb all common shocks for banks in a group. We define the groups according to banks' (i) capital structure and (ii) exposure to the financial crisis. First, poorly capitalized banks face tighter constraints and are more affected by new regulations than well-capitalized banks. Similarly, a low capital ratio may point to severe losses during the crisis. Such banks may cut back loans and increase interest rates more strongly. At the same time, their tax exposure is *ceteris paribus* larger as equity is exempt. To control for the impact of the capital structure and of shocks that differently affect banks depending on their capitalization, we add *capital ratio-specific* year fixed effects. We interact a dummy variable for each quartile of the capital ratio (equity/total assets) distribution in the full sample with the year fixed effect. Using the distribution in 2009 ensures that the groups are defined independently of any later changes in the capital structure possibly induced by the levy. Second, the financial crisis affected countries differently. Ireland, for example, was severely hit, whereas some eastern European countries like Poland or Slovakia did not experience a banking crisis at all. Whenever countries that were severely affected are more likely to adopt a levy, the estimates can be biased. Country-year fixed effects resolve this problem but would absorb all country-level variation in the levy variables. The effects would be estimated only by exploiting the variation within those

¹⁰ For example, consider a bank with large equity losses during the financial crisis. It will deleverage, thereby raising the lending rate. At the same time, the smaller equity renders such a bank more exposed to the levy (as equity is exempt) based on a pre-introduction but post-crisis balance sheet (e.g., 2009, 2010). This might still bias the estimates although the predicted tax exposure is undistorted by any behavioral responses to the levy itself.

four countries that exempt some banks. We estimate such a variant as a robustness check (see Online Appendix B.1.2); for the main analysis, we instead define three groups of countries depending on how strongly they were hit by the financial crisis and include separate year fixed effects. We measure the crisis exposure by the cumulative fiscal costs of recapitalization and asset relief during the sample period (as a share of 2013 GDP). We include *crisis-specific* year fixed effects and interact the year dummy with a dummy variable for countries that experience (i) no banking crisis, (ii) a crisis with fiscal costs smaller than the median conditional on having incurred a positive cost (Luxembourg: 5.7% of GDP), and (iii) a severe crisis with costs larger than the median. This approach identifies the effect based on a comparison of banks with the same approximate capital structure in countries similarly affected by the crisis.

We control for several bank- and country-level characteristics in all regressions. The bank-level covariates are size measured by the logarithm of total assets, $Assets_{ijt-1}$, and a proxy for the cost structure, $Cost_{ijt}$, measured by the non-interest expenditures divided by total assets. The stock variable $Assets$ is included with a lag because it is observed at the end of the year. We also control for portfolio risk: Since measures like the average regulatory risk weight or the non-performing loans ratio are only reported by a few banks, we include the loan loss provisions, $Provisions_{ijt}$. Negative or positive but small values point to good loan quality. The capital structure is indirectly taken into account by the capital ratio-specific year fixed effects. Noting the findings of Devereux et al. (2015), the capital ratio itself would be endogenous because banks adjust their leverage in response to the levy. At the country level, we include interbank rate $Interbank_{jt}$, statutory corporate income tax rate CIT_{jt} , and a proxy for the number of banks, the Herfindahl–Hirschman index of market concentration HHI_{jt} . The real growth rate of GDP, $Growth_{jt}$, and inflation, $Inflation_{jt}$, account for macroeconomic conditions, and the annual fiscal costs of recapitalization and asset relief as a share of GDP, $Fiscal\ Cost_{jt}$, capture the contemporaneous effect of government interventions during the crisis.

Compared to the difference-in-difference methodology applied by Buch et al. (2016) and Capelle-Blancard and Havrylchyk (2017), the fixed-effects estimator is more suitable for cross-country data because the levy was introduced at different points in time and different tax rates apply. Apart from the endogeneity of the bank-level levy variables that requires instruments, a concern is that banks might have anticipated the tax. Such a scenario is likely under two circumstances: (i) a lengthy political process before the levy is effective or (ii) an earlier introduction in other countries. The former can usually be ruled out as, for example, the time between legislation and enactment was between one and three months only [see Devereux et al. (2015, Appendix)]. The latter is a possibility because a levy was introduced in Sweden already in 2009 and in Hungary in 2010. However, the IMF report that established a benchmark was issued in September 2010 and most countries adopted the levy shortly afterward with a tax design that differs from the Hungarian levy. Although anticipation is possible in three countries that adopted the levy in 2012, a robustness test shows that excluding their banks from the sample does not alter the empirical results (see Online Appendix B.1.4).

4.1.2 Heterogeneity in responses

Theory implies that the main adjustment margin differs between banks depending on their initial balance sheet composition, that is, whether they borrow from or lend to the money market. Noting the balance sheet identity, one can express the condition for the main scenario, $m_i \geq 0$, in terms of the ratio of loans to customer deposits for which an empirical counterpart exists: $l_i/(d_i + s_i) \geq 1/(1 - k)$. Banks with high values are loan-rich and deposit-poor and finance loans partly by non-deposit liabilities; they are expected to respond according to the main scenario and to raise lending and deposit rates. In contrast, those with low values have substantial asset holdings apart from loans; they are expected to adjust according to the alternative scenario and tend to lower the deposit rate. To test this prediction, we run main regression (12) separately for banks with a high and low loan-to-deposit ratio (i.e., above or below 100%), respectively.

A second prediction is that the pass-through to borrowers is reinforced by a high degree of market concentration and weakened by a high capital ratio, which reflects tight capital standards. We include interaction terms between the tax variables and measures of competition BC_j and capitalization CAP_{ij} :

$$y_{ijt} = \alpha_i + \gamma_t + \beta_1 \text{Levy}_{ijt} + \beta_2 \text{Levy}_{ijt} * BC_j + \varphi X_{ijt} + \varepsilon_{ijt}, \quad (13)$$

$$y_{ijt} = \alpha_i + \gamma_t + \beta_1 \text{Levy}_{ijt} + \beta_2 \text{Levy}_{ijt} * CAP_{ij} + \varphi X_{ijt} + \varepsilon_{ijt}. \quad (14)$$

Since theory highlights the importance of the number of competitors and the interest rate elasticity, we represent BC_j by the Herfindahl–Hirschman index, which equals the sum of bank market shares squared. We use the HHI based on assets of the entire banking industry in a country provided by the ECB. Furthermore, we use the number of bank branches per 10,000 inhabitants (branch density): It can be interpreted as a proxy for the switching cost and thus for the demand elasticity. The capital structure CAP_{ij} , in turn, is represented by the regulatory capital ratio (regulatory capital/risk-weighted assets) and the capital ratio (equity/total assets). All concentration and capitalization variables are pre-introduction (2009) values and independent of any later changes induced by the levy.¹¹

4.2 Data and measurement

We employ an unbalanced panel dataset with bank-level information of (at most) 2776 banks in 23 European countries between 2007 and 2013. All 27 EU member states except for France, Hungary, Slovenia, and Finland, which adopted conceptually different levies, are included in the analysis. The sample consists of 18,146 bank-year observations. Four different types of banks are represented: commercial banks (17.8% of all bank-year observations), savings banks (23.6%), cooperative banks (54.8%), and real estate and mortgage banks (3.9%). The sample covers approx-

¹¹ Therefore, the main effects cannot be estimated because they are absorbed by the bank fixed effect. The contemporaneous effect is captured by the standard control variable HHI; if the branch density is used in (13), we replace the contemporaneous control variable accordingly.

imately 44% of all banks, and the composition by country is shown in Table 14 (Appendix A.2).

The main source is *Bankscope*, a database provided by the Bureau van Dijk with financial information from banks' annual reports. Since taxes are levied on single entity accounts, we rely on unconsolidated financial statements. This alleviates the double taxation problem of multinational banks as we have separate data for the parent bank and its subsidiaries although foreign branches can be subject to a levy abroad and are part of the unconsolidated account of the domestic bank. Detailed information about bank levies is taken from Devereux et al. (2015). Macroeconomic data are from Eurostat, banking sector characteristics from the ECB Banking Structures Report, and interbank rates from the OECD financial statistics or national central banks.¹² Some adjustments of the sample are made: Observations with closing date between April and September are excluded because they cannot be clearly attributed to a specific year; observations with closing date between January and March (October and December) are attributed to the previous (current) year. All observations are transformed into Euro, and inactive banks as well as banks with negative assets or equity are deleted.¹³ Eventually, some variables are ratios defined in terms of assets, loans, or deposits and may take extreme values if, for example, assets are very small or misreported. We reduce the influence of such outliers by winsorizing all bank-level ratios at the 2.5 and 97.5 level.

To define the bank-level tax variables, we approximate the taxable liabilities of each bank in those four countries that exempt small banks or apply a progressive tax rate—Austria, Germany, the Netherlands, and the UK—using the information provided by Devereux et al. (2015, Appendix). The taxable liabilities usually equal total liabilities net of customer deposits protected by a guarantee scheme; in Germany, all customer deposits are exempt.¹⁴

The bank-year dummy variable *Levy2* is equal to one if both a levy is charged in a year and the taxable liabilities of a bank exceed the allowance threshold. In the seven other countries with a levy but no allowance, all banks are taxed. Hence, *Levy2* is the same for all banks and coincides with the country-year variable *Levy1*.

¹² Information from central banks for Bulgaria, Cyprus, Latvia, Lithuania, Malta, and Romania

¹³ This might give rise to survivorship bias if banks exit because they cannot pass through the levy. However, this is highly unlikely in the short run as most countries introduced the levy not earlier than 2011 and the sample includes just two follow-up periods.

¹⁴ The taxable liabilities are approximated as follows:

- AT: Total liabilities—insured deposits
- DE: Total liabilities—customer deposits
- NL: Total liabilities and equity—insured deposits—regulatory capital, if regulatory capital not reported, it is replaced by common equity or total equity
- UK: Total liabilities and equity—insured deposits—Tier 1 equity; if Tier 1 equity is not reported, it is replaced by common equity or total equity

We use the current-year balance sheet, only for Austria, we use the 2010 balance sheet.

Insured deposits are computed by multiplying a bank's customer deposits by the coverage ratio, which equals the volume share of insured deposits at the country level. Data on the coverage ratio are from the Joint Research Centre of the European Commission and are available for the years 2007, 2011, 2012 (reports 'JRC Report under Article 12 of Directive 94/19/EC as amended by Directive 2009/14/EC,' 2010, and 'Updated Estimates for EU eligible and covered deposits,' 2014). For the years 2010 and 2013, we use the values of the next or previous year.

The variable *Levy3* equals the statutory marginal tax rate in a given year. We select the marginal tax rate that applies for each bank according to its taxable liabilities in Austria and Germany, which have a progressive tax schedule. All other countries apply a uniform tax rate, and *Levy3* is the same for all banks that are taxed. Table 15 (Appendix A.2) summarizes detailed information about how the two tax variables are assigned at the bank level. The instruments for *Levy2* and *Levy3* rely on the predicted tax exposure. They are defined in exactly the same way but according to the taxable liabilities calculated for a bank's 2007 balance sheet.

In general, the bank-level tax variables are more informative than their country-level counterpart, but they hinge on the approximation of the tax base. Data limitations prevent us from taking all features of the levy into account. For example, we do not have any data on insured deposits at the bank level and approximate them by multiplying a bank's customer deposits with the volume share of insured deposits at the country level (coverage ratio) on which some data are available. The Netherlands and the UK only exempt specific equity positions (regulatory or Tier 1 capital), which are reported by few banks. If not available, we instead deduct total equity, which is typically larger, and might understate the tax exposure. Those two countries also charge the levy on banks' consolidated balance sheets: We compute the bank-level tax exposure according to the consolidated balance sheet and match it back to the unconsolidated data on interest rates and bank characteristics used in the subsequent analysis. This approach is consistent with national tax law and keeps the data comparable across countries. However, it considerably restricts the sample as banks must report both consolidated and unconsolidated information on all variables of interest. Consequently, British and Dutch banks are underrepresented in the full sample. A robustness test which only uses consolidated data of these banks and does not match the tax exposure back to the unconsolidated statement leads to similar results as the baseline approach (Online Appendix B.1.2). Finally, we cannot account for any effects of a levy that is also imposed on derivatives like in Austria, Germany, and Portugal. However, this aspect seems unlikely to influence the tax incidence on borrowers and depositors.

4.3 Descriptive evidence

The summary statistics are shown in Table 16 (Appendix A.2). On average, total assets of a bank amount to EUR 5.43bn, and the capital ratio is 8.67%. Banks located in levy countries are smaller and funded by a lower share of equity (7.79% vs. 10.75%). Banks effectively taxed are, in turn, considerably larger (total assets EUR 13.43bn vs. 3.27bn) and have a lower capital ratio (7.92% vs. 8.87%). This difference reflects a typical feature of the levy: the exemption of small banks (allowance) and of equity. It is taken into account by controlling for size in all specifications and by including capital ratio-specific year fixed effects.

Roughly 70% of banks are located in 11 out of 23 countries that adopt a levy. The tax exposure and the marginal tax rates are shown in Table 3: In 2010, the levy was

Table 3 Exposure of banks to levy

	Banks in levy country		Banks subject to levy		Mean tax rate	Total banks
	Fraction	Banks	Fraction	Banks		
2010	2.69	74	2.70	74	0.018	2749
2011	69.57	1876	19.55	516	0.030	2734
2012	70.90	1999	19.68	529	0.036	2711
2013	68.21	1294	22.94	437	0.038	1922

The first two columns refer to banks in a country that charges a bank levy (i.e., Levy1 = 1); the next two columns refer to banks that are effectively taxed (i.e., Levy2 = 1); the fifth column states the mean marginal tax rate (in % of taxable liabilities) conditional on a bank being taxed (i.e., Levy2 = 1)

first effective in Sweden¹⁵ and applied to all 74 banks. In 2011, the share of banks in countries that charge a levy strongly increased to 70% as seven countries including Germany and the UK introduced the tax. However, only 20% were effectively taxed due to the exemption of small banks. These shares slightly increased in 2012 when three additional countries adopted a levy. On average, the marginal tax rate computed for banks which are subject to the levy is 0.032%. It ranges between 0.018 (Sweden, 2010) and 0.4% (Slovakia).

Online Appendix B.2 provides estimates of the levy's actual fiscal cost. On average, the tax is charged on a base that represents more than half of a bank's total liabilities and is larger than the customer deposits. We calculate mean fiscal costs between EUR 2.1m and EUR 3.6m per bank and year. This roughly equals a quarter of a bank's annual tax expense and 10% of its pre-tax profit. Due to allowance and progressive tax rates, the fiscal cost is negligible for small banks but rises more than proportionately with bank size.

Banks earn a mean interest income on loans of 5.86% of loans. Banks in countries that adopt the levy at one point in time show higher values (6.04%) than those in countries without (5.42%). Figure 1 illustrates the mean interest income as well as the lower and upper quartiles of the distribution (left panel): It steadily decreased from more than 7% in 2008 to less than 5% in 2013, which reflects a general decline of interest rates. The right panel illustrates the mean interest income depending on whether a bank becomes subject to the levy at one point. In the pre-introduction period, one observes a similar pattern for both groups. In 2011 when seven countries adopted a levy, the interest income of taxed banks increased, whereas that of unaffected banks fell.

The mean interest expense on customer deposits equals 1.67% of customer deposits. Higher values are reported by banks in countries that adopt a levy (2.18% vs. 1.54%) and by banks that are effectively taxed (2.17% vs. 1.57%). One can attribute this difference to Italian banks that are not subject to a levy and are characterized by a low interest expense on deposits (1.23%). Without Italy, banks in countries with no levy pay a higher interest expense (2.42%). A robustness test in Online Appendix B.1.4

¹⁵ Officially, it was enacted in October 2009 and formally introduced on 30 December, and the first payment was already due in 2009. Following Devereux et al. (2015), banks are considered unaffected in 2009 because many had drawn up the balance sheet when it was enacted.

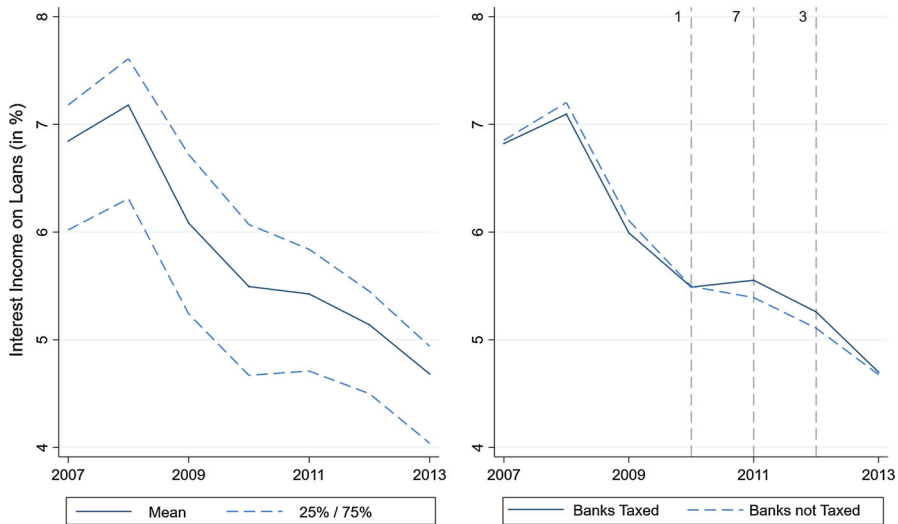


Fig. 1 Interest income on loans. The left panel illustrates the mean interest income on loans/avg. loans and the upper and lower quartile. The right panel shows the mean int. income depending on whether banks are taxed (i.e., $\text{Levy2} = 1$) at least once during the sample period. The vertical lines indicate the introduction of the levy in 2010 (one country), 2011 (7 countries), and 2012 (3 countries)

demonstrates that the levy has an even stronger effect once we exclude Italian banks. Figure 2 shows how the interest expense on deposits evolves over time. The left panel reveals a declining pattern that stabilized after 2010. In the right panel, one can observe a comparable trend prior to the introduction and a relatively stronger increase in 2011 if banks are taxed.

Data on the interest expense are available only for a subsample of 924 banks. Most Austrian and German banks do not report this indicator, and those two countries, which account for almost two-thirds of all bank-year observations, are underrepresented in this subsample (see Table 14). Compared to the full sample, banks which report the interest expense on deposits have a slightly smaller interest income on loans and net interest margin, larger assets (EUR 7.13bn) and twice as large loan loss provisions (EUR 38.2m). We take this into account by controlling for these factors. Online Appendix B.1.4 further explores this aspect and estimates the effect of the levy on the other two outcome variables (i.e., IIL and NIM) restricting the sample to banks that report the interest expense on deposits at least once during the observation period. The effects on interest income on loans are broadly consistent with those obtained in the full sample, but the effects on the net interest margin become insignificant.

The average net interest margin equals 2.49%. It is lower for banks in countries that adopt a levy (2.41% vs. 2.66%) and that are effectively taxed (2.18% vs. 2.57%). Compared to the interest rates, it only slightly declined as illustrated in Fig. 3. The right panel shows the mean depending on the tax exposure: In 2011, there was a relative increase in the NIM of taxed banks; subsequently, it did not decline more strongly.

The descriptive evidence reveals a positive association of the bank levy and the interest rates and margins at the time of its introduction especially in 2011, which is

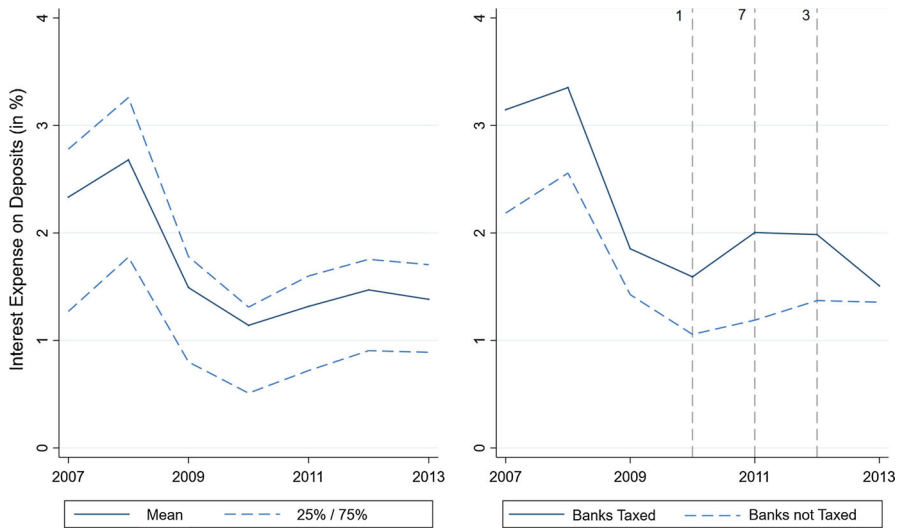


Fig. 2 Interest expense on deposits. The left panel illustrates the mean interest expense on customer deposits/avg. customer deposits and the upper and lower quartile. The right panel shows the mean int. expense of banks depending on whether they are taxed (i.e., $\text{Levy2} = 1$) at least once during the sample period. The vertical lines indicate the introduction years

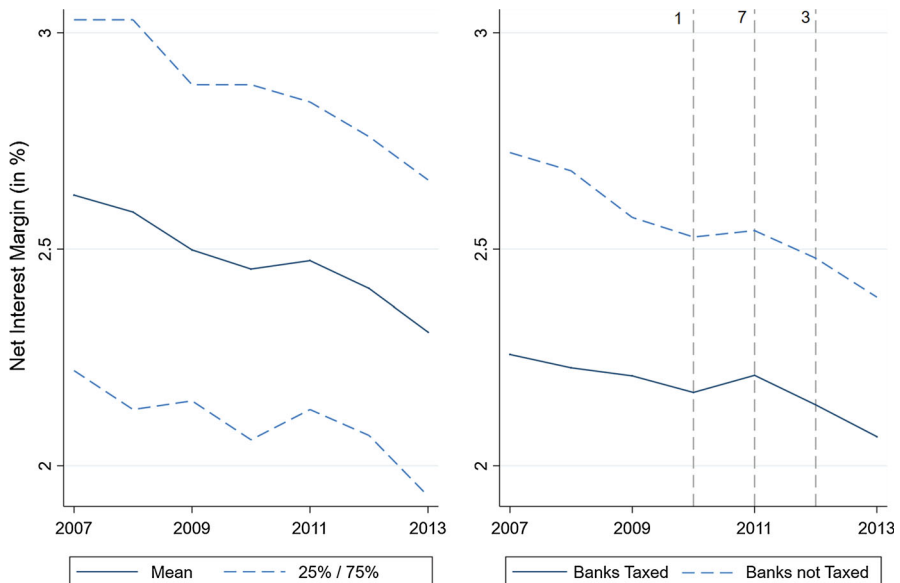


Fig. 3 Net interest margin. The left panel illustrates the mean net interest margin and the upper and lower quartile. The right panel shows the mean net interest margin of banks depending on whether they are taxed (i.e., $\text{Levy2} = 1$) at least once during the sample period. The vertical lines indicate the introduction years

in line with theory. Furthermore, tax and outcome variables markedly differ between banks with different capital ratios and between countries differently affected by the

financial crisis (measured by the fiscal outlays). Online Appendix B.3 provides more detailed statistics (Fig. B.1–B.4). In particular, weakly capitalized banks are taxed more often and report higher interest rates. Similarly, banks in countries with positive fiscal outlays are more likely to be taxed and tend to raise deposit rate and net interest margin by more than those in countries without. These correlations motivate separate time trends that absorb all group-specific variation.

4.4 Main results

This section summarizes the estimation results from the baseline and the heterogeneous response models. Our findings are robust as documented in a series of robustness tests in Online Appendix B.1 that control for additional bank- and country-level shocks, use consolidated accounts to calculate banks' tax exposure, and evaluate the estimation results in various subsamples.¹⁶

4.4.1 Baseline model

Tables 4, 5 and 6 report the coefficients of the baseline specification (12). For each of the three outcome variables, we run eight regressions representing the tax exposure of banks by the country-year and bank-year dummy variables *Levy1* and *Levy2* in Columns 1–2 and 3–5 as well as by the marginal tax rate *Levy3* in Columns 6–8. In each case, we report the OLS and IV estimates when using the bank-level tax exposure and the estimates from regressions with bank and year fixed effects as well as with bank and group-year fixed effects, respectively.

The bank levy raises the interest income on loans once a bank is effectively taxed as shown in Table 4: The coefficient of *Levy2* (Cols. 3–5) points to a significant increase in the average lending rate that ranges from 0.18 to 0.25 percentage points. The linear and quadratic terms of the marginal tax rate *Levy3* are jointly significant as indicated by the *F* test and positive: Evaluating the marginal effect $\beta_1 + 2\beta'_1 \cdot \tau$ at the median,¹⁷ $\tau = 0.02\%$, yields values between 5.71 and 6.68. Therefore, raising the marginal tax rate from the median (Germany, banks with taxable liabilities between EUR 200m and EUR 10bn) toward the 75th percentile value (Sweden, Latvia), $\tau = 0.036\%$, increases the interest income by between 0.09 and 0.11 percentage points as stated in the row 'marg. effect.' These effects are small relative to the average interest income of 5.86%.

According to the estimates in Table 5, the levy also increases the interest expense on customer deposits. This points to the substitution effect that arises because deposits are largely exempt. Compared to the average of 1.67%, the increase is not negligible: A bank located in a country with a levy pays an average interest rate that is between 0.19 and 0.33 percentage points higher. For banks effectively taxed (Cols. 3–5), the increase is between 0.16 and 0.21 percentage points. The coefficients of the marginal tax rate remain insignificant (Cols. 6–8), however.

¹⁶ For example, we obtain comparable estimates when excluding countries that adopt the levy after 2011 to address anticipation concerns; excluding the largest country from the sample yields qualitatively similar estimates for IIL and IED but insignificant ones for NIM (see Table B.4).

¹⁷ Conditional that banks are effectively taxed, i.e., $\text{Levy2} = 1$

Table 4 Baseline regressions: interest income on loans

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Levy _{ijt}	0.018 (0.061)	−0.038 (0.064)	0.182*** (0.037)	0.253*** (0.047)	0.238*** (0.051)	6.141*** (2.012)	7.112*** (2.348)	5.947*** (2.298)
Levy _{ijt} ²						−8.234 (6.349)	−10.830 (7.025)	−5.666 (7.141)
Assets _{ijt−1}	0.066 (0.129)	0.031 (0.131)	0.077 (0.130)	0.089 (0.134)	0.049 (0.136)	0.072 (0.129)	0.080 (0.133)	0.048 (0.136)
Cost _{ijt}	0.121*** (0.021)	0.120*** (0.021)	0.123*** (0.021)	0.125*** (0.022)	0.126*** (0.022)	0.123*** (0.021)	0.125*** (0.022)	0.126*** (0.022)
Provisions _{ijt}	−0.000** (0.000)	−0.000* (0.000)	−0.000** (0.000)	−0.000* (0.000)	−0.000 (0.000)	−0.000* (0.000)	−0.000* (0.000)	−0.000 (0.000)
Inflation _{jt}	−0.081*** (0.024)	−0.092*** (0.027)	−0.078*** (0.023)	−0.076*** (0.023)	−0.087*** (0.027)	−0.072*** (0.023)	−0.069*** (0.023)	−0.092*** (0.026)
Growth _{jt}	−0.005 (0.012)	−0.015 (0.016)	−0.012 (0.012)	−0.015 (0.012)	−0.028** (0.015)	−0.012 (0.011)	−0.014 (0.011)	−0.030** (0.014)
Interbank _{jt}	0.109*** (0.032)	0.125*** (0.031)	0.100*** (0.030)	0.096*** (0.031)	0.107*** (0.030)	0.105*** (0.031)	0.105*** (0.031)	0.111*** (0.030)
CIT _{jt}	−1.333 (1.780)	−0.644 (1.841)	−1.830 (1.743)	−1.595 (1.789)	−1.239 (1.836)	−2.790* (1.617)	−2.380 (1.673)	−2.460 (1.777)
Fiscal Cost _{jt}	0.143*** (0.030)	0.108*** (0.029)	0.143*** (0.024)	0.142*** (0.024)	0.110*** (0.026)	0.136*** (0.023)	0.135*** (0.023)	0.106*** (0.026)
HHI _{jt}	9.878*** (2.827)	−6.269* (3.370)	12.498*** (2.673)	13.362*** (2.731)	−2.046 (3.245)	13.554*** (2.644)	13.742*** (2.719)	−2.105 (3.391)
Marg. effect						0.093	0.107	0.092
F test (<i>p</i> value)						0.001	0.002	0.005
Obs.	13,332	13,332	13,314	13,120	13,120	13,314	13,120	13,120
No. banks	2438	2438	2435	2386	2386	2435	2386	2386
R ²	0.633	0.655	0.637	0.637	0.660	0.638	0.638	0.662
Levy	Levy1	Levy1	Levy2	Levy2	Levy2	Levy3	Levy3	Levy3
Method	OLS	OLS	OLS	IV	IV	OLS	IV	IV
Bank FE	YES	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	NO	YES	YES	NO	YES	YES	NO
Year × cap. ratio FE	NO	YES	NO	NO	YES	NO	NO	YES
Year × crisis FE	NO	YES	NO	NO	YES	NO	NO	YES

Dep. variable: interest income on loans/av. loans; tax exposure represented by country-year dummy Levy1 in (1)–(2), bank-year dummy Levy2 in (3)–(5), and marginal tax rate Levy3 in (6)–(8); robust standard errors clustered at the bank level in parentheses: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Marg. effect equals the marginal effect $[\beta_1 + 2\beta'_1 \tau] \cdot d\tau$ of raising the marginal tax rate from the median $\tau = 0.02\%$ by $d\tau = 0.016\%$ to the 75th percentile value. *F* test is the *p* value of the test whether the marginal effect is zero

Table 5 Baseline regressions: interest expense on customer deposits

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Levy _{ijt}	0.189*** (0.073)	0.329*** (0.084)	0.164** (0.080)	0.172** (0.082)	0.214*** (0.083)	0.935 (2.045)	1.072 (2.075)	1.657 (2.074)
Levy _{ijt} ²						− 3.174 (5.117)	− 3.559 (5.190)	− 2.565 (5.270)
Assets _{ijt−1}	0.472*** (0.107)	0.429*** (0.104)	0.464*** (0.107)	0.466*** (0.107)	0.430*** (0.105)	0.465*** (0.108)	0.467*** (0.108)	0.439*** (0.106)
Cost _{ijt}	0.026 (0.018)	0.017 (0.018)	0.027 (0.018)	0.026 (0.018)	0.016 (0.018)	0.023 (0.018)	0.023 (0.019)	0.015 (0.018)
Provisions _{ijt}	− 0.000*** (0.000)	− 0.000* (0.000)	− 0.000*** (0.000)	− 0.000*** (0.000)	− 0.000** (0.000)	− 0.000*** (0.000)	− 0.000*** (0.000)	− 0.000** (0.000)
Inflation _{jt}	− 0.016 (0.022)	− 0.015 (0.023)	− 0.015 (0.022)	− 0.016 (0.022)	− 0.016 (0.024)	− 0.017 (0.023)	− 0.018 (0.023)	− 0.019 (0.025)
Growth _{jt}	− 0.090*** (0.011)	− 0.096*** (0.013)	− 0.089*** (0.011)	− 0.089*** (0.011)	− 0.090*** (0.013)	− 0.082*** (0.011)	− 0.082*** (0.011)	− 0.082*** (0.013)
Interbank _{jt}	0.054*** (0.024)	0.031 (0.024)	0.054** (0.024)	0.055** (0.025)	0.038 (0.024)	0.065*** (0.023)	0.066*** (0.023)	0.050*** (0.022)
CIT _{jt}	7.148*** (1.243)	6.842*** (1.286)	6.282*** (1.264)	6.559*** (1.194)	6.543*** (1.319)	6.894*** (1.281)	7.186*** (1.209)	6.848*** (1.314)
Fiscal Cost _{jt}	0.051*** (0.015)	0.076*** (0.015)	0.047*** (0.014)	0.048*** (0.014)	0.071*** (0.015)	0.047*** (0.014)	0.048*** (0.014)	0.071*** (0.015)
HHI _{jt}	3.090 (3.227)	7.396* (3.916)	2.888 (3.328)	2.985 (3.349)	5.292 (3.881)	0.308 (3.055)	0.324 (3.064)	2.438 (3.624)
Marg. effect						0.013	0.015	0.025
F test (<i>p</i> value)						0.661	0.619	0.405
Obs.	4665	4665	4661	4645	4645	4661	4645	4645
No. banks	874	874	873	872	872	873	872	872
R ²	0.570	0.594	0.570	0.572	0.594	0.569	0.571	0.593
Levy	Levy1	Levy1	Levy2	Levy2	Levy2	Levy2	Levy3	Levy3
Method	OLS	OLS	OLS	IV	IV	OLS	IV	IV
Bank FE	YES	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	NO	YES	YES	NO	YES	YES	NO
Year × cap. ratio FE	NO	YES	NO	NO	YES	NO	NO	YES
Year × crisis FE	NO	YES	NO	NO	YES	NO	NO	YES

Dep. variable: interest expense on customer deposits/av. customer deposits; tax exposure represented by country-year dummy Levy1 in (1)–(2), bank-year dummy Levy2 in (3)–(5), and marginal tax rate Levy3 in (6)–(8); robust standard errors clustered at the bank level in parentheses: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Marg. effect equals the marginal effect $[\beta_1 + 2\beta'_1\tau] \cdot d\tau$ of raising the marginal tax rate from the median $\tau = 0.02\%$ by $d\tau = 0.016\%$ to the 75th percentile value. *F* test is the *p* value of the test whether the marginal effect is zero

The coefficients in Table 6 reveal a higher net interest margin (NIM): In countries that adopt a levy, it rises in the range of 0.11 and 0.21 percentage points. The coefficients of Levy2 imply a smaller increase between 0.05 and 0.09 percentage points. The coefficients of the marginal tax rate are jointly significant: Evaluating them at the median, one obtains marginal effects in a range of 1.54 and 2.51. Increasing the tax rate from the median to the 75th percentile value thus leads to a between 0.025 and 0.04 percentage points higher NIM. Comparing the estimates to the average NIM of 2.48% suggests that the latter increases by at most 3.6% once a bank is taxed. The NIM is less sensitive than the interest rates because the rising deposit rate weakens the increase in the lending spread. This measure also includes returns on other assets and liabilities that are likely less responsive to a levy.

Instrumenting the actual by the predicted bank-level tax exposure (Cols. 4–5 and 7–8) consistently leads to larger coefficient estimates in all regressions. A potential explanation is that banks adjust their balance sheets, for instance, by downsizing, deleveraging, or substituting insured deposits in order to reduce the marginal tax rate if the tax schedule is progressive or to avoid paying the levy altogether if there is an allowance. When defining the tax variable based on their current balance sheet, we classify such banks as not taxed or taxed at a lower rate. Nevertheless, their behavioral responses may affect aggregate loans and deposits, thereby increasing interest income and expense. It is, therefore, plausible that the OLS estimates understate the true difference between treatment and control group.

Overall, the bank levy is shifted to borrowers and raises the interest rate spread, while depositors may benefit from higher interest rates. The quantitative effects are moderate especially for lending rate and net interest margin. One can partly attribute this to the use of averages as outcome variables. The estimates are broadly consistent with the previous literature: A positive effect on deposit rates is also found by Buch et al. (2016) for Germany, but they do not find any significant increases in lending rates. However, we can show that the levy's effect on interest income on loans is weaker or even insignificant if it is estimated for German banks only (see Online Appendix B.1.5). For Hungary, Capelle-Blancard and Havrylchyk (2017) estimated stronger effects on lending rate and net interest margin. However, the Hungarian bank levy is the highest in Europe, and the banks in the sample have significant market power due to long-lasting borrowing relationships. For the corporate income tax, in turn, Albertazzi and Gambacorta (2010) estimated that a 10 percentage points decrease in the tax rate lowers the pre-tax profit by 9.4%. This is stronger than our largest estimated effect on the NIM (the most closely related outcome) of 3.6%. In comparison with capital regulation, the levy does not raise the NIM by more than a one percentage point increase in capital requirements, which is associated with a 0.14 percentage points higher interest rate spread in the Euro area (Slovik and Cournède 2011).

4.4.2 Heterogeneity: balance sheet composition

The empirical findings—the levy increases lending and deposit rates and net interest margin—support the main scenario of the Monti–Klein model. Theory predicts substantial differences in the adjustment of banks depending on their balance sheet composition, more precisely, on their initial loan-to-deposit ratio. To explore this

Table 6 Baseline regressions: net interest margin

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Levy _{ijt}	0.206*** (0.020)	0.112*** (0.025)	0.062*** (0.016)	0.088*** (0.019)	0.048*** (0.020)	2.263*** (0.754)	2.751*** (0.845)	1.604** (0.816)
Levy _{ijt} ²						− 4.571*** (2.014)	− 5.823*** (2.229)	− 1.624 (2.171)
Assets _{ijt−1}	− 0.253*** (0.050)	− 0.250*** (0.046)	− 0.304*** (0.049)	− 0.301*** (0.049)	− 0.286*** (0.045)	− 0.307*** (0.049)	− 0.305*** (0.049)	− 0.287*** (0.045)
Cost _{ijt}	0.082*** (0.011)	0.089*** (0.011)	0.062*** (0.010)	0.062*** (0.010)	0.077*** (0.010)	0.062*** (0.010)	0.061*** (0.010)	0.077*** (0.010)
Provisions _{ijt}	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000* (0.000)	0.000 (0.000)	0.000 (0.000)	0.000* (0.000)
Inflation _{jt}	0.041*** (0.010)	0.026*** (0.010)	0.053*** (0.010)	0.053*** (0.010)	0.031*** (0.010)	0.054*** (0.010)	0.055*** (0.010)	0.030*** (0.010)
Growth _{jt}	0.006 (0.005)	0.002 (0.006)	0.017*** (0.005)	0.016*** (0.005)	0.008 (0.005)	0.017*** (0.005)	0.017*** (0.005)	0.007 (0.005)
Interbank _{jt}	0.034*** (0.013)	0.036*** (0.012)	0.051*** (0.013)	0.049*** (0.013)	0.044*** (0.012)	0.053*** (0.013)	0.053*** (0.012)	0.044*** (0.011)
CIT _{jt}	− 1.850* (0.977)	− 3.459*** (0.962)	− 1.442 (0.922)	− 1.462 (0.943)	− 3.536*** (0.929)	− 1.563* (0.927)	− 1.534 (0.949)	− 3.860*** (0.930)
Fiscal Cost _{jt}	0.020*** (0.005)	0.007 (0.005)	0.002 (0.004)	0.001 (0.004)	− 0.004 (0.005)	0.000 (0.004)	− 0.001 (0.004)	− 0.005 (0.005)
HHI _{jt}	5.716*** (1.011)	2.270*** (0.913)	4.971*** (0.928)	5.245*** (0.936)	1.849** (0.884)	5.213*** (0.971)	5.390*** (0.979)	2.020** (0.925)
Marg. effect						0.033	0.040	0.025
F test (<i>p</i> value)						0.002	0.001	0.036
Obs.	14,962	14,962	14,922	14,673	14,673	14,922	14,673	14,673
No. banks	2714	2714	2713	2652	2652	2713	2652	2652
R ²	0.123	0.246	0.111	0.110	0.247	0.112	0.111	0.248
Levy	Levy1	Levy1	Levy2	Levy2	Levy2	Levy3	Levy3	Levy3
Method	OLS	OLS	OLS	IV	IV	OLS	IV	IV
Bank FE	YES	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	NO	YES	YES	NO	YES	YES	NO
Year × cap. ratio FE	NO	YES	NO	NO	YES	NO	NO	YES
Year × crisis FE	NO	YES	NO	NO	YES	NO	NO	YES

Dep. variable: net interest margin; tax exposure represented by country-year dummy Levy1 in (1)–(2), bank-year dummy Levy2 in (3)–(5), and marginal tax rate Levy3 in (6)–(8); robust standard errors clustered at the bank level in parentheses: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Marg. effect equals the marginal effect $[\beta_1 + 2\beta'_1 \tau] \cdot d\tau$ of raising the marginal tax rate from the median $\tau = 0.02\%$ by $d\tau = 0.016\%$ to the 75th percentile value. *F* test is the *p* value of the test whether the marginal effect is zero

aspect further, we run the main regression in two subsamples of banks with a high and low loan-to-deposit ratio prior to the first introduction of bank levies (2009). The sample is split at the cutoff $l_i/(d_i + s_i) = 100\%$ close to the theoretically motivated value $l_i/(d_i + s_i) = 1/(1 - k)$.¹⁸

The results reveal striking differences: Table 7 reports the coefficients of interest obtained in the subsample of loan-rich and deposit-poor banks with a loan-to-deposit ratio of at least 100%. In this case, the bank levy significantly raises the average lending and deposit rate as well as the net interest margin. These findings are in line with the theoretical prediction of the main scenario (see Table 2). The effects are comparable to those estimated in Sect. 4.4.1. In addition, the country-year dummy *Levy1* now significantly raises the interest income on loans. For interest expense and net interest margin, one observes slightly larger increases than in the full sample.

As shown in Table 8, the response of loan-poor and deposit-rich banks to the levy is much weaker and more ambiguous. We find no consistent evidence that such banks raise lending and deposit rates. In contrast, the estimates hint at the possibility of a lower deposit rate as predicted in the alternative scenario but the coefficients are significant only when using the marginal tax rate (Cols. 6–8). In line with theory, the net interest margin rises, but the effect is rather small and significant in only half of the specifications.

The contrasting effects are consistent with the prediction that the balance sheet composition determines the banks' adjustment mechanism. Especially, the main scenario derived from the model can rationalize the empirical evidence for deposit-poor, loan-rich banks. These banks tend to drive the results for the full sample.

4.4.3 Heterogeneity: market concentration and capital structure

Another key prediction is that the magnitude of the pass-through to borrowers depends on market concentration and bank capitalization. We estimate the levy's effect on interest income on loans and run regressions (13) and (14), thereby interacting the tax variables with measures of competition and capital structure.

Table 9 reports the coefficients of interest with bank competition being represented by the Herfindahl–Hirschman index and by the branch density (i.e., no. bank branches/10,000 inhabitants). The interaction terms are positive and significant in seven out of eight cases if the HHI is used as a measure of competition (upper panel) and negative and significant in all cases if the branch density is used (lower panel). Therefore, the increase in the average lending rate and the pass-through to borrowers are more pronounced in concentrated banking markets.

A quantitative interpretation requires the marginal effects: Table 10 summarizes the increase in interest income on loans if

1. a bank is located in a country that charges a levy (i.e., $\text{Levy1} = 1$): $\beta_1 + \beta_2 \cdot \text{BC}$ with $\text{BC} = \{\text{HHI}, \text{BD}\}$;
2. a bank is effectively taxed (i.e., $\text{Levy2} = 1$): $\beta_1 + \beta_2 \cdot \text{BC}$;

¹⁸ The capital requirement k is defined in terms of risk-weighted assets, on which data are not available for all banks in the sample. Given the small value of k especially if expressed in terms of total assets, 100% appears a suitable approximation that is only slightly smaller than the theoretical cutoff.

Table 7 Heterogeneous responses: deposit-poor banks (loans/customer deposits > 100%)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Interest income on loans</i>								
<i>Levy_{ijt}</i>	0.237*** (0.068)	0.150* (0.078)	0.203*** (0.061)	0.230*** (0.070)	0.149* (0.081)	6.814** (3.020)	7.282** (3.219)	5.074* (3.020)
<i>Levy_{ijt}²</i>						-16.137** (7.287)	-17.324** (7.798)	-7.440 (7.565)
Marg. effect						0.089	0.105	0.076
<i>F</i> test						0.024	0.024	0.080
Obs.	4277	4277	4277	4262	4262	4277	4262	4262
No. banks	780	780	780	775	775	780	775	775
<i>R</i> ²	0.702	0.735	0.701	0.701	0.735	0.701	0.701	0.736
<i>Interest expense on deposits</i>								
<i>Levy_{ijt}</i>	0.393*** (0.109)	0.537*** (0.136)	0.408*** (0.110)		0.558*** (0.136)	7.810*** (2.821)		10.907*** (3.367)
<i>Levy_{ijt}²</i>						-21.269*** (6.910)		-26.784*** (8.585)
Marg. effect						0.111		0.157
<i>F</i> test (<i>p</i> value)						0.006		0.001
Obs.	2925	2925	2925		2925	2925		2925
No. banks	539	539	539		539	539		539
<i>R</i> ²	0.631	0.665	0.631		0.666	0.630		0.664
<i>Net interest margin</i>								
<i>Levy_{ijt}</i>	0.241*** (0.032)	0.173*** (0.035)	0.146*** (0.029)	0.166*** (0.033)	0.093*** (0.033)	2.935*** (1.107)	3.007*** (1.171)	1.171 (1.147)

Table 7 continued

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$Levy2_{ijt}^2$						-5.794* (3.119)	-6.008* (3.273)	0.4230 (3.229)
Marg. effect						0.043	0.044	0.018
F test (p value)						0.007	0.009	0.251
Obs.	4780	4780	4780	4748	4748	4780	4748	4748
No. banks	856	856	856	846	846	856	846	846
R^2	0.290	0.386	0.278	0.278	0.380	0.275	0.275	0.380
Levy	Levy1	Levy1	Levy1	Levy2	Levy2	Levy3	Levy3	Levy3
Method	OLS	OLS	OLS	OLS	IV	OLS	IV	IV
Bank FE	YES	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	NO	YES	YES	NO	YES	YES	NO
Year $\times$ cap. ratio FE	NO	YES	NO	NO	YES	NO	NO	YES
Year $\times$ crisis FE	NO	YES	NO	NO	YES	NO	NO	YES

Dep. variables: interest income on loans/av. loans (upper panel); interest expense on customer deposits/av. customer deposits (second panel), net interest margin (lower panel); clustered standard errors in parentheses: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. All regressions include the full set of bank- and country-level control variables

For interest expense on deposits (second panel), the estimates in Columns 5 and 8 are obtained using OLS because the instrument fully predicts the instrumented variable in the corresponding subsample

Marg. effect equals the marginal effect $[\beta_1 + 2\beta_1'\tau] \cdot d\tau$ of raising the marginal tax rate from the median $\tau = 0.02\%$ by $d\tau = 0.016\%$ to the 75th percentile value. F test is the p value of the test whether the marginal effect is zero

Table 8 Heterogeneous responses: deposit-rich banks (loans/customer deposits < 100%)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Interest income on loans</i>								
<i>Levy_{ijt}</i>	-0.042 (0.085)	-0.087 (0.092)	0.084** (0.042)	0.156*** (0.054)	0.164*** (0.056)	2.616 (2.449)	3.472 (2.948)	3.289 (2.932)
<i>Levy_{ijt}²</i>						-2.239 (6.224)	-4.617 (7.375)	-1.477 (7.850)
Marg. effect						0.040	0.057	0.052
<i>F</i> test (<i>p</i> value)						0.253	0.217	0.220
Obs.	8729	8729	8725	8558	8558	8725	8558	8558
No. banks	1592	1592	1591	1548	1548	1591	1548	1548
<i>R</i> ²	0.659	0.667	0.659	0.660	0.668	0.659	0.661	0.669
<i>Interest expense on deposits</i>								
<i>Levy_{ijt}</i>	0.107 (0.103)	0.161 (0.117)	-0.009 (0.106)	0.018 (0.113)	-0.011 (0.114)	-4.573** (2.278)	-4.106* (2.398)	-4.144* (2.420)
<i>Levy_{ijt}²</i>						12.017** (5.783)	10.741* (6.076)	12.138** (6.174)
Marg. effect						-0.065	-0.059	-0.059
<i>F</i> test (<i>p</i> value)						0.046	0.089	0.093
Obs.	1642	1642	1638	1625	1625	1638	1625	1625
No. banks	314	314	313	312	312	313	312	312
<i>R</i> ²	0.556	0.586	0.555	0.560	0.591	0.558	0.562	0.594
<i>Net interest margin</i>								
<i>Levy_{ijt}</i>	0.081** (0.031)	-0.006 (0.040)	0.020 (0.017)	0.039* (0.022)	0.024 (0.022)	1.583 (1.042)	2.192* (1.186)	1.794 (1.129)

Table 8 continued

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Levy _{ijt} ²						-3.338 (2.747)	-4.897 (3.087)	-2.579 (2.977)
Marg. effect						0.023	0.032	0.027
<i>F</i> test (<i>p</i> value)						0.123	0.062	0.098
Obs.	9745	9745	9737	9537	9537	9737	9537	9537
No. banks	1772	1772	1771	1722	1722	1771	1722	1722
<i>R</i> ²	0.104	0.190	0.104	0.106	0.195	0.105	0.107	0.196
Levy	Levy1	Levy1	Levy1	Levy2	Levy2	Levy3	Levy3	Levy3
Method	OLS	OLS	OLS	IV	IV	OLS	IV	IV
Bank FE	YES	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	NO	YES	YES	NO	YES	YES	NO
Year × cap. ratio FE	NO	YES	NO	NO	YES	NO	NO	YES
Year × crisis FE	NO	YES	NO	NO	YES	NO	NO	YES

Dep. variables: interest income on loans/av. loans (upper panel); interest expense on customer deposits/av. customer deposits (second panel), net interest margin (lower panel); clustered standard errors in parentheses: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. All regressions include the full set of bank- and country-level control variables

Marg. effect equals the marginal effect $[\beta_1 + 2\beta_1\tau] \cdot d\tau$ of raising the marginal tax rate from the median $\tau = 0.02\%$ by $d\tau = 0.016\%$ to the 75th percentile value. *F* test is the *p* value of the test whether the marginal effect is zero

Table 9 Heterogeneous responses: interest income on loans and market concentration

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Levy</i> $\times$ <i>HHI</i>								
<i>Levy</i> _{ijt}	−0.232*** (0.081)	−0.377*** (0.088)	−0.066 (0.060)	0.016 (0.074)	0.012 (0.079)	0.821 (2.445)	2.497 (3.300)	2.530 (3.491)
<i>Levy</i> _{ijt} ²						−25.345** (11.730)	−23.813** (11.786)	−15.400 (12.390)
<i>Levy</i> _{ijt} * <i>HHI</i> _j	8.128*** (1.626)	9.232*** (1.876)	7.129*** (1.654)	6.278*** (1.722)	5.853*** (1.875)	95.257** (43.157)	76.825 (47.563)	57.304 (52.007)
Obs.	13,332	13,332	13,314	13,120	13,120	13,314	13,120	13,120
No. banks	2438	2438	2435	2386	2386	2435	2386	2386
<i>R</i> ²	0.636	0.659	0.639	0.640	0.662	0.638	0.639	0.662
<i>Levy</i> $\times$ <i>branch density</i>								
<i>Levy</i> _{ijt}	0.653*** (0.170)	0.801*** (0.195)	0.601*** (0.166)	1.200** (0.503)	1.653*** (0.509)	9.446*** (3.644)	9.694** (4.071)	11.325*** (4.092)
<i>Levy</i> _{ijt} ²						−12.223 (7.837)	−13.615 (8.700)	−13.173 (8.951)
<i>Levy</i> _{ijt} * <i>BD</i> _j	−0.147*** (0.036)	−0.186*** (0.042)	−0.103*** (0.035)	−0.227** (0.111)	−0.321*** (0.112)	−0.947** (0.454)	−0.844* (0.462)	−1.048** (0.504)
Obs.	13,332	13,332	13,314	13,120	13,120	13,314	13,120	13,120
No. banks	2438	2438	2435	2386	2386	2435	2386	2386
<i>R</i> ²	0.634	0.658	0.637	0.637	0.661	0.637	0.638	0.663
<i>Levy</i>	<i>Levy</i> 1	<i>Levy</i> 1	<i>Levy</i> 2	<i>Levy</i> 2	<i>Levy</i> 2	<i>Levy</i> 3	<i>Levy</i> 3	<i>Levy</i> 3
Method	OLS	OLS	OLS	IV	IV	OLS	IV	IV
Bank FE	YES	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	NO	YES	YES	NO	YES	YES	NO
Year $\times$ cap. ratio FE	NO	YES	NO	NO	YES	NO	NO	YES
Year $\times$ crisis FE	NO	YES	NO	NO	YES	NO	NO	YES

Dep. variable: interest income on loans; the levy variables are interacted with the 2009 HHI (upper panel) and branch density (lower panel); clustered standard errors in parentheses: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. All regressions include the full set of bank- and country-level control variables

If the branch density is used as a competition measure, the (contemporaneous) branch density replaces the HHI in the set of controls

3. and the marginal tax rate is increased from the median $\tau = 0.02\%$ by $d\tau = 0.016\%$ toward the 75th percentile: $[\beta_1 + 2\beta'_1 \cdot \tau + \beta_2 \cdot BC] \cdot d\tau$.

In each case, we rely on the most conservative estimates from (IV) regressions with group-year fixed effects (i.e., Cols. 2, 5, 8 of Table 9). Since we explore cross-country heterogeneity, we evaluate the effects at the 25th, 50th, and 75th percentile of the distribution with all countries equally weighted.¹⁹ As a comparison, we also add the effect at the sample median.

¹⁹ The full sample distribution is less meaningful for a cross-country comparison as fragmented markets are overrepresented due to the large number of German banks.

Table 10 Marginal effects: magnitude of pass-through and competition

			(1)	(2)	(3)
<i>HHI</i>	25th percentile	UK	0.059	0.287***	0.074**
	Median	Cyprus	0.627***	0.647***	0.130**
	75th percentile	Slovakia	0.709***	0.699***	0.138**
	Sample Median	Germany	− 0.183***	0.133**	0.050
<i>Branch Density</i>	25th percentile	Slovakia	0.374***	0.519***	0.134***
	Median	Romania	0.213**	0.402***	0.119***
	75th percentile	Austria	− 0.134*	0.159***	0.088**
	Sample Median	Germany	− 0.083	0.183***	0.093**

This table reports the marginal effects of the bank levy on interest income on loans. The levy is represented by the country-year dummy (Col. 1), bank-year dummy (Col. 2), and the marginal tax rate that is raised from the median to p75 (Col. 3). The percentiles are based on the respective 2009 distribution with countries equally weighted. We also report the marginal effects evaluated at the median of the distribution with countries being weighted according to their representation in the full sample

The pass-through is stronger in concentrated markets: Whenever a dummy variable is interacted with the HHI, the interest income on loans increases in the range of 0.6 to 0.7 percentage points in highly concentrated markets (Slovakia, HHI = 0.127; Cyprus, 0.109). Increasing the marginal tax rate from median to the 75th percentile still raises the interest income by roughly 0.13 percentage points, which is stronger than in the baseline specification (0.09 pp). In fragmented markets like the UK (0.047), the increase is weaker but still significant in two cases. Using the branch density as a proxy for bank competition, one observes increases between 0.37 and 0.52 percentage points in countries at the 25th percentile (Slovakia, 2.29 branches/10,000 inhabitants) and between 0.21 and 0.4 percentage points at the median (Romania, 3.16). At the 75th percentile (Austria, 5.03), these effects become more ambiguous. The branch density also influences the effects of a 0.016 percentage points higher marginal tax rate although the differences are less pronounced. Evaluating the heterogeneous response model at the weighted sample median (Germany, HHI = 0.021, 4.76 branches/10,000 inhabitants) yields estimates broadly consistent with the baseline model.

Overall, the pass-through in highly concentrated markets with increases up to 0.7 percentage points is economically significant compared to the mean interest income on loans of 5.86%. Figure 4 illustrates the pass-through of a bank that is subject to the levy depending on the HHI (Col. 2). The blue points indicate the quartiles and the sample median. The increase in the lending rate is significant already at low degrees of concentration and rapidly increases in the HHI.

Table 11 shows the coefficients of the interaction terms between the tax variables and two proxies of the capital structure, namely the 2009 regulatory capital ratio (upper panel) and the ratio of equity to total assets (lower panel). The results are somewhat ambiguous: The levy increases the average lending rate of banks with a high regulatory capital less strongly. This result supports the prediction that the adjustment of well-capitalized banks is less pronounced. However, we find no consistent evidence that a high capital ratio weakens the pass-through.

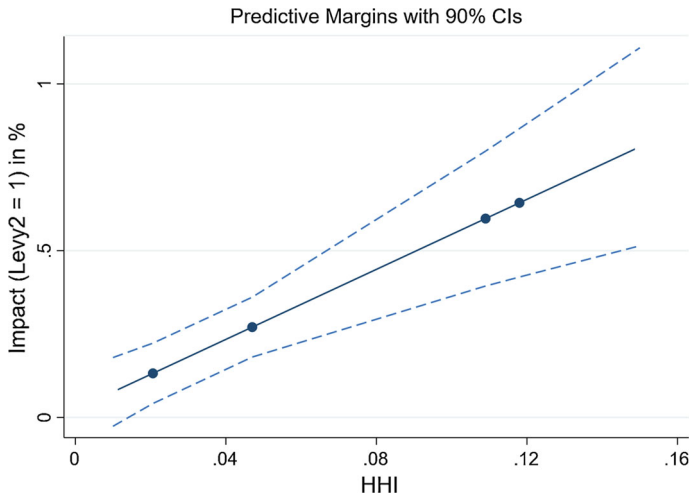


Fig. 4 Heterogeneity: interest income on loans and HHI. This figure illustrates the increase in IIL if a bank is taxed ($\text{Levy2} = 1$) depending on the HHI. The points indicate sample median (0.021) and quartiles (0.047, 0.107, 0.127) of the distribution with equally weighted countries

Table 12 summarizes the marginal effect of the levy depending on the regulatory capital ratio. We evaluate the same three scenarios, that is, introducing a levy at country and bank level and raising the marginal tax rate from the median to the 75th percentile value, at the 25th, 50th, and 75th percentile of the 2009 distribution of the regulatory capital ratio. We also report the effect at the regulatory minimum of 8%, but very few banks have such a low capital ratio. The analysis again relies on the most conservative estimates (i.e., Cols. 2, 5, 8 of Table 11).

These results suggest that mainly banks with a low regulatory capital ratio shift the levy to borrowers. First, the marginal effects of the country-year tax variable on interest income on loans (Col. 1) are typically insignificant as in the baseline specification. Second, the bank-year dummy in Column 2 causes a significant and diminishing increase in the average lending rate between 0.1 and 0.45 percentage points as long as the regulatory capital ratio does not exceed the median (15.2%). If banks are better capitalized, the pass-through vanishes. Figure 5 illustrates this scenario. Third, raising the marginal tax rate from the median by 0.016% (Col. 3) only induces banks with a very low regulatory capital ratio close to the minimum to significantly raise lending rates. Due to the nonlinearities, the same tax increase will influence the behavior of more banks if implemented in a country with an already high level of tax rates (e.g., at the 75th percentile, $\tau = 0.036\%$), however.

5 Conclusion

In the aftermath of the financial crisis, several European countries introduced levies on bank liabilities to raise a contribution from the financial sector and to cover the fiscal costs of future banking crises. Whether the incidence of this novel tax is on banks is a

Table 11 Heterogeneous responses: interest income on loans and capital structure

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Levy × regulatory capital ratio</i>								
Levy _{ijt}	0.287*** (0.107)	0.208* (0.112)	0.487*** (0.177)	0.858*** (0.227)	0.837*** (0.2227)	6.802** (3.414)	9.107** (4.220)	9.032** (4.135)
Levy _{ijt} ²						3.069 (7.735)	1.168 (9.207)	2.306 (9.319)
Levy _{ijt} *RCAP _{ij}	−0.019*** (0.005)	−0.019*** (0.005)	−0.027** (3.414)	−0.049*** (4.220)	−0.049*** (4.135)	−0.325* (0.188)	−0.408* (0.226)	−0.398* (0.207)
Obs.	10,285	10,285	10,276	10,129	10,129	10,276	10,129	10,129
No. banks	1866	1866	1866	1825	1825	1866	1825	1825
R ²	0.676	0.702	0.676	0.676	0.701	0.676	0.676	0.702
<i>Levy × capital ratio</i>								
Levy _{ijt}	−0.025 (0.070)	−0.094 (0.079)	0.085 (0.071)	0.101 (0.081)	0.121 (0.086)	6.596*** (2.017)	7.484*** (2.379)	7.407*** (2.405)
Levy _{ijt} ²						−8.135 (6.482)	−10.774 (7.109)	−5.546 (7.083)
Levy _{ijt} *CAP _{ij}	0.006 (0.004)	0.006 (0.005)	0.014 (0.010)	0.023** (0.011)	0.016 (0.012)	−0.054 (0.184)	−0.043 (0.190)	−0.165 (0.181)
Obs.	13,332	13,332	13,314	13,120	13,120	13,314	13,120	13,120
No. banks	2438	2438	2435	2386	2386	2435	2386	2386
R ²	0.633	0.656	0.637	0.637	0.660	0.638	0.638	0.662
Levy	Levy1	Levy1	Levy2	Levy2	Levy2	Levy3	Levy3	Levy3
Method	OLS	OLS	OLS	IV	IV	OLS	IV	IV
Bank FE	YES	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	NO	YES	YES	NO	YES	YES	NO
Year × cap. ratio FE	NO	YES	NO	NO	YES	NO	NO	YES
Year × crisis FE	NO	YES	NO	NO	YES	NO	NO	YES

Dep. variable: interest income on loans; the levy variables are interacted with the 2009 regulatory capital ratio (upper panel) and capital ratio (lower panel); clustered standard errors in parentheses: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. All regressions include the full set of bank- and country-level control variables

priori unclear as they may pass it onto their customers. The main prediction derived from the Monti–Klein model is that banks shift the tax to borrowers. In addition, they have an incentive to attract more deposits, which are partly exempt, such that depositors earn a higher interest rate. Evidence from 2776 banks in 23 EU countries (2007–2013) shows that the bank levy raises the average lending and deposit rate as well as the net interest margin. Whenever a bank is taxed, the interest income on loans (as a share of average loans) and the interest expense on deposits (as a share of average deposits) increase by more than 0.2 percentage points and the net interest margin by 0.05 percentage points. The effects are small relative to the mean of 5.86 and 2.49%. They are mainly driven by the behavior of loan-rich, deposit-

Table 12 Magnitude of pass-through and capital structure

		(1)	(2)	(3)
<i>Regulatory capital ratio</i>				
Regulatory minimum	8%	0.056	0.449***	0.095**
25th percentile	12.76%	−0.036	0.218***	0.065
Median	15.2%	−0.083	0.100*	0.049
75th percentile	18.57%	−0.149**	−0.064	0.027

This table reports the marginal effects of the bank levy on interest income on loans. The levy is represented by the country-year dummy (Col. 1), bank-year dummy (Col. 2), and the marginal tax rate that is raised from the median to p75 (Col. 3). The percentiles are based on 2009 distribution of the regulatory capital ratio

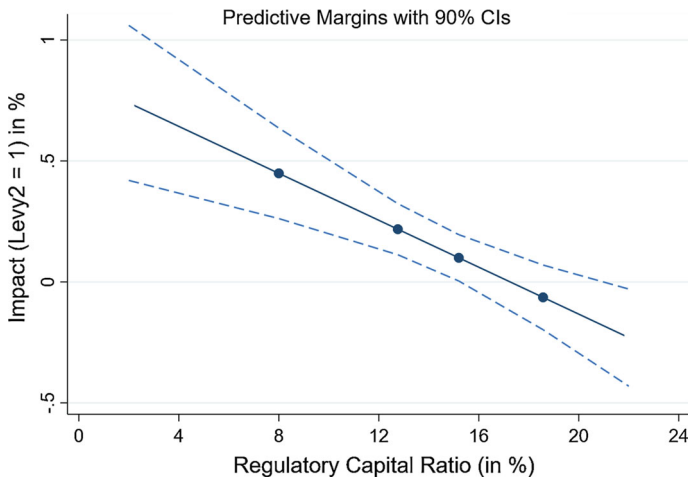


Fig. 5 Heterogeneity: interest income on loans and regulatory capital ratio. This figure illustrates the effect of a bank being taxed ($\text{Levy2} = 1$) on IIL depending on the regulatory capital ratio (col. 2 in Table 12). The points indicate capital ratio at regulatory minimum 8% and the corresponding percentiles (12.76%, 15.2%, 18.57%)

poor banks, which is a novel result consistent with theory. Market concentration and capital structure influence the extent to which the levy is passed onto borrowers: In concentrated markets, the average lending rate increase is significantly stronger. There is also evidence for a stronger pass-through by banks with a low regulatory capital ratio.

This paper adds to the literature cross-country evidence about the incidence of bank levies. It generalizes previous insights obtained at the country level and sheds light on the potential effects of banks' contributions to the new Single Resolution Fund in the Euro area established in 2016. The combination of theory and evidence identifies and rationalizes the (heterogeneous) responses of banks. We show that especially balance sheet composition, market concentration, and capital structure matter and find considerable differences.

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A Appendix

A.1 Proofs

Proof of Proposition 1 We differentiate the first-order conditions with respect to loans and insured deposits in (4). For loans L , we get

$$\frac{\partial L}{\partial \tau} = \frac{1 - k}{r'_L(L) \left(1 - \frac{1}{N\varepsilon_L}\right)} < 0. \quad (\text{A.1})$$

Noting $r_L = r_L(L)$ and $r'_L(L) < 0$, the lending rate increases (see (5)). The effect of N , ε_L and k is negative by inspection and reduces the increase in r_L .

Differentiating the first-order condition with respect to insured deposits S yields

$$\frac{\partial S}{\partial \tau} = \frac{1}{r'_S(S) \left(1 + \frac{1}{N\varepsilon_S}\right)} > 0. \quad (\text{A.2})$$

Again, noting $r_S = r_S(S)$ and $r'_S(S) > 0$, we get (6). The effect of N and ε_S on r_S is positive and reinforces the increase in S and r_S . $\square$

Proof of Proposition 2 To get the profit maximization problem (7), we substitute $T_i = \tau d_i$, $e_i = k(l_i - m_i)$ and $l_i + m_i = s_i + d_i + e_i$ into (1). The second and third equation jointly imply $m_i = l_i - (s_i + d_i)/(1 - k)$ and $e_i = k(s_i + d_i)/(1 - k)$.

$$\begin{aligned} \pi_i = & (1 + r_L)l_i - (1 + r) \left[l_i - \frac{s_i + d_i}{1 - k} \right] - (1 + r_S)s_i \\ & - (1 + r_D)d_i - (1 + \rho) \frac{k(s_i + d_i)}{1 - k}. \end{aligned} \quad (\text{A.3})$$

By rearranging and using the definition $\tilde{r}$, one obtains equation (7).

We differentiate the first-order condition for uninsured deposits D in (8):

$$\frac{\partial D}{\partial \tau} = -\frac{1}{r'_L(L) \left(1 + \frac{1}{N\varepsilon_D}\right)} < 0. \quad (\text{A.4})$$

Using $r_D = r_D(D)$ with $r'_D(D) > 0$ yields (9). The effect of N and ε_D on r_D is negative by inspection and reinforces the effect. $\square$

Proof of Proposition 3 In the main scenario $m_i \geq 0$, the NIM defined by the first part of (11) responds to the levy τ according to:

$$\frac{\partial NIM}{\partial \tau} = \frac{\partial r_L}{\partial \tau} - \frac{(r - r_D)D + (r - r_S)S}{L^2} \frac{\partial L}{\partial \tau} + \frac{r - r_S - r'_S(S)S}{L} \frac{\partial S}{\partial \tau}. \quad (\text{A.5})$$

We substitute (5) as well as (A.1) and (A.2) for the partial derivatives of lending rate, loans and insured deposits and use $r - r_S = r_S/(N\varepsilon_S) - \tau$ implied by the third equation of (4). We evaluate this response at an initial tax rate $\tau = 0$ to capture the effect of introducing the levy:

$$\begin{aligned} \frac{\partial NIM}{\partial \tau} &= \frac{1-k}{1-\frac{1}{N\varepsilon_L}} \left[1 - \frac{(r-r_D)D + (r-r_S)S}{r'_L(L)L^2} \right] - \frac{1}{1+\frac{1}{N\varepsilon_S}} \left(1 - \frac{1}{N} \right) \frac{S}{L} \\ &= \frac{1-k}{1-\frac{1}{N\varepsilon_L}} \left[1 + \frac{\varepsilon_L}{r_L} \frac{(r-r_D)D + (r-r_S)S}{L} \right] - \frac{1}{1+\frac{1}{N\varepsilon_S}} \left(1 - \frac{1}{N} \right) \frac{S}{L}. \end{aligned} \quad (\text{A.6})$$

Noting $S = L - D - M - E$ such that $S/L = 1 - k - (D + M)/L < 1 - k$, the effect is positive.

In the alternative scenario $m_i < 0$, the NIM defined by the second part of (11) responds according to

$$\begin{aligned} \frac{\partial NIM}{\partial \tau} &= - \frac{(1-k)[(r-r_L)L + (\tilde{r}-r_D)D + (\tilde{r}-r_S)S]}{(D+S)^2} \frac{\partial D}{\partial \tau} \\ &\quad + \frac{(1-k)[\tilde{r}-r_D-r'_D(D)D]}{D+S} \frac{\partial D}{\partial \tau}. \end{aligned} \quad (\text{A.7})$$

Using the definition of NIM as well as $\tilde{r} - r_D = r_D/(N\varepsilon_D) - \tau$ implied by (8), we evaluate this effect at $\tau = 0$:

$$\frac{\partial NIM}{\partial \tau} = - \left[NIM + \frac{(1-k)r_D}{\varepsilon_D} \left(1 - \frac{1}{N} \right) \right] \frac{1}{D+S} \frac{\partial D}{\partial \tau} > 0. \quad (\text{A.8})$$

Noting $\partial D/\partial \tau < 0$ in (9), introducing a levy unambiguously raises the NIM. $\square$

A.2 Data information, measurement, and summary statistics

See Tables 13, 14, 15, 16.

Table 13 Definition of variables

Variable	Description	Source
IIL	Int. income on loans/avg. loans	Bankscope
IED	Int. expense on customer deposits/avg. customer deposits	Bankscope
NIM	Net interest margin	Bankscope
Levy1	Levy dummy at country level	Devereux et al. (2015)
Levy2	Levy dummy at bank level	Author's calculations
Levy3	Marginal tax rate at bank level	Author's calculations
Assets	(Log of) Total assets	Bankscope
BD	Bank branches per 10,000 inhabitants	ECB
CIT	Corporate income tax rate	Devereux et al. (2015)
Cost	Non-interest/total assets	Bankscope
Equity	Equity/total assets (capital ratio)	Bankscope
Fiscal Cost	Fiscal costs of recapitalization and asset relief (% of GDP)	European Commission (2013)
Growth	Growth rate of real GDP	ECB
HHI	Herfindahl–Hirschman index (assets)	ECB
Inflation	Inflation rate	ECB
Interbank	Interbank rate (Euro area: 3M EURIBOR)	OECD, Central Banks
Provisions	Loan loss provisions	Bankscope
Reg Cap	Regulatory capital ratio	Bankscope

Table 14 Sample composition by Country. *Source:* ECB Monetary and Financial Statistics

	Banks 2010		Sample 2010		Sample 2010 (IED)	
	No.	Share	No.	Share	No.	Share
11 Countries adopt levy 2010–2013						
<i>Austria</i>	750	11.58	208	7.57	6	0.66
<i>Belgium</i>	48	0.74	26	0.95	16	1.75
<i>Cyprus</i>	127	1.96	7	0.26	7	0.77
<i>Germany</i>	1819	28.08	1540	56.02	20	2.19
<i>Latvia</i>	29	0.45	16	0.58	16	1.75
<i>Netherlands</i>	254	3.92	2	0.07	2	0.22
<i>Portugal</i>	133	2.05	18	0.65	15	1.64
<i>Romania</i>	33	0.51	13	0.47	13	1.42
<i>Slovakia</i>	15	0.23	8	0.29	7	0.77
<i>Sweden</i>	148	2.28	74	2.69	70	7.67
<i>UK</i>	239	3.69	33	1.20	24	2.63
12 Countries do not adopt levy 2010–2013						
<i>Bulgaria</i>	24	0.37	17	0.62	16	1.75
<i>Czech Republic</i>	36	0.56	19	0.69	18	1.97
<i>Denmark</i>	143	2.21	79	2.88	71	7.78
<i>Estonia</i>	7	0.11	3	0.11	2	0.22
<i>Greece</i>	36	0.56	9	0.33	8	0.88
<i>Ireland</i>	461	7.12	3	0.11	2	0.22
<i>Italy</i>	697	10.76	498	18.12	496	54.33
<i>Lithuania</i>	77	1.19	7	0.26	7	0.77
<i>Luxembourg</i>	118	1.82	47	1.71	37	4.05
<i>Malta</i>	26	0.40	6	0.22	6	0.66
<i>Poland</i>	685	10.57	29	1.06	26	2.85
<i>Spain</i>	255	3.94	87	3.17	44	4.82
Total	6160		2749		913	

This table compares the number of active banks to those in the sample in 2010. The last two columns refer to banks that report the interest expense on deposits at least once

Table 15 Bank-level tax exposure. *Sources:* OECD (2013); Devereux et al. (2015, Appendix)

	Levy2	Levy3	Taxable Liabilities	Time Period
<i>Austria</i>	0	0%	$T \leq \text{EUR } 1\text{bn}$	2011–2013
	1	0.055%	$\text{EUR } 20\text{bn} \geq T > \text{EUR } 1\text{bn}$	
	1	0.085%	$T > \text{EUR } 20\text{bn}$	
<i>Belgium</i>	1	0.035%	all banks	2012–2013
<i>Cyprus</i>	1	0.095%	all banks	2011–2012
	1	0.011%	all banks	2013
<i>Germany</i>	0	0%	$T \leq \text{EUR } 0.2\text{bn}$	2011–2013
	1	0.02%	$\text{EUR } 10\text{bn} \geq T > \text{EUR } 0.2\text{bn}$	
	1	0.03%	$\text{EUR } 100\text{bn} \geq T > \text{EUR } 10\text{bn}$	
	1	0.04%	$\text{EUR } 200\text{bn} \geq T > \text{EUR } 100\text{bn}$	
	1	0.05%	$\text{EUR } 300\text{bn} \geq T > \text{EUR } 200\text{bn}$	
	1	0.06%	$T \geq \text{EUR } 300\text{bn}$	
<i>Latvia</i>	1	0.036%	all banks	2011–2012
	1	0.072%	all banks	2013
<i>Netherlands</i>	0	0%	$T \leq \text{EUR } 20\text{bn}$	2012–2013
	1	0.033%	$T > \text{EUR } 20\text{bn}$	2012–2013
<i>Portugal</i>	1	0.05%	all banks	2011–2013
<i>Romania</i>	1	0.1%	all banks	2011–2013
<i>Slovakia</i>	1	0.4%	all banks	2012–2013
<i>Sweden</i>	1	0.018%	all banks	2010
	1	0.036%	all banks	2011–2013
<i>UK</i>	0	0%	$T \leq \text{GBP } 20\text{bn}$	2011–2013
	1	0.05625%	$T > \text{GBP } 20\text{bn}$	2011
	1	0.066%	$T > \text{GBP } 20\text{bn}$	2012
	1	0.0975%	$T > \text{GBP } 20\text{bn}$	2013

The second column states the bank-year dummy Levy2 potentially depending on taxable liabilities T . The third column states the marginal tax rate: In Austria and Germany with a progressive tax schedule, we select the tax rate that applies for a bank given T . In the Netherlands and the UK, which have a tax rate independent of bank size but exempt small banks, the Levy3 is the same for all taxed banks with T larger than the allowance. We use a simple average of the tax rate on short- and long-term borrowing. In all other countries with a uniform tax rate and no allowance, it is the same for all banks

Table 16 Summary statistics

	Full sample			No levy			With levy			Banks not taxed			Banks taxed				
	Mean	SD	Obs.	Min	Max	Mean	SD	Obs.	Mean	SD	Obs.	Mean	SD	Obs.			
III	5.86	1.68	15,638	3.06	11.43	5.42	1.91	4581	6.04	1.53	11,057	5.86	1.62	12,253	5.86	1.87	3385
IED	1.67	1.24	5440	0.34	6.03	1.54	1.16	4387	2.20	1.43	1053	1.57	1.18	4577	2.17	1.46	863
NIM	2.49	0.82	18,146	0.52	4.70	2.66	1.02	5397	2.41	0.71	12,749	2.57	0.78	14,297	2.18	0.88	3885
Assets	5427	43,355	18,146	0.32	2,246,381	7547	36,011	5397	4529	46,323	12,749	3273	22,434	14,297	13,429	83,579	3849
Equity	8.67	4.28	18,146	2.94	24.08	10.74	4.62	5397	7.79	3.80	12,749	8.87	4.15	14,297	7.92	4.68	3849
Cost	2.65	1.25	18,137	0.26	7.01	2.94	1.35	5389	2.52	1.18	12,748	2.74	1.20	14,289	2.30	1.38	3849
Provisions	18.25	169.38	17,704	-1041	10,139	41.33	286.44	5180	8.70	79.46	12,524	16.18	175.81	13,955	25.93	142.69	3749
Inflation	2.05	1.19	18,146	-1.70	15.30	2.34	1.31	5397	1.93	1.11	12,749						
Fiscal Cost	0.68	1.14	18,146	0.00	24.60	0.36	1.33	5397	0.82	1.01	12,749						
Growth	0.61	3.04	18,146	-17.70	10.50	-0.50	2.83	5397	1.08	3.00	12,749						
CIT	29.28	4.41	18,146	10.00	38.90	27.18	4.12	5397	30.17	4.23	12,749						
HHI	0.04	0.03	18,146	0.02	0.34	0.05	0.03	5397	0.03	0.02	12,749						
BD	4.83	1.22	18,146	1.10	12.33	5.44	1.69	5397	4.58	0.82	12,749						

In percent; assets and provisions in EUR (million), the HHI is normalized in the unit interval. The second and third columns show the statistics for banks in countries that do (not) adopt a levy during the sample period (i.e., Levy1 = 1). The fourth and fifth columns show the statistics for banks that are (not) taxed during the sample period (i.e., Levy2 = 1)

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